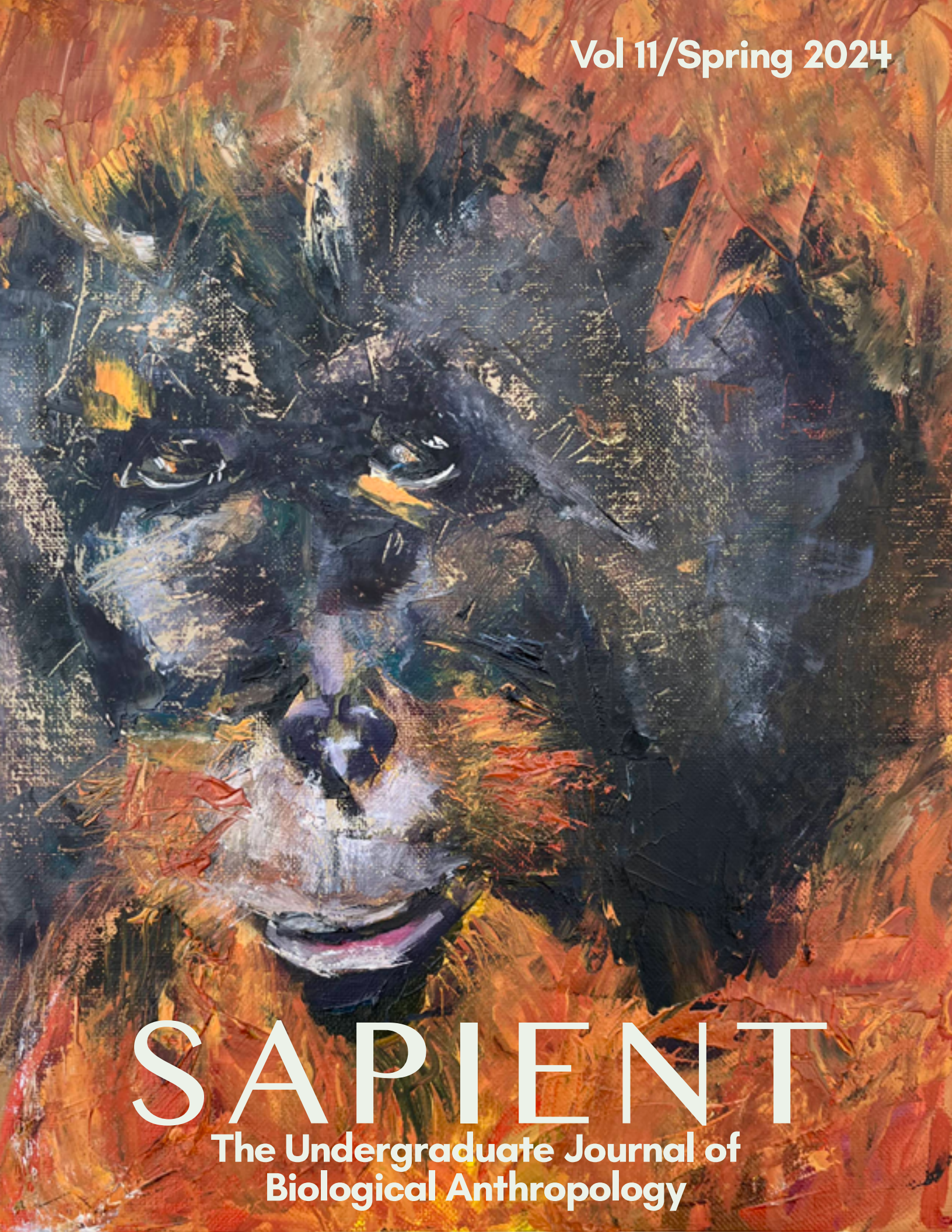


An abstract painting of a human face, rendered in a highly textured, expressive style. The face is composed of various shades of brown, tan, and grey, with dark, swirling lines around the eyes and mouth. The background is a mix of warm, earthy tones like orange, red, and yellow, with some cooler blue and green accents. The overall effect is one of intense, raw emotion and a sense of ancient, primal humanity.

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SAPIENT

The Undergraduate Journal of
Biological Anthropology

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Anthropology
Spring 2024

Sapient is Columbia University's peer-reviewed undergraduate journal of biological anthropology. Founded in 2012 to give undergraduates at Columbia a place to share and reflect upon bioanthropological works of academic and artistic merit, *Sapient* has continued to develop. For questions, comments, or to get involved visit sapientjournal.com.

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Letter from the Editors

Dear Reader,

From the Editorial board, we are proud to present Volume 11 of *Sapient*: Columbia's Undergraduate Journal of Biological Anthropology. We are excited to showcase a wide array of exceptional student art, poetry, and of course, academic writing. This year's volume includes papers on a wide variety of topics, including primate growth and development, paleolithic art, and evolutionary theory.

We are grateful to have had so many new faces in *Sapient* meetings and events this year, each bringing a new skill set. We also began working with our first illustrators in recent years. We are glad to have had submissions from board editors, as well as those that came from outside Columbia and the department of E3B. Thank you for helping us create a multimedia, interdisciplinary celebration of undergraduate work.

Needless to say, it was a long and tumultuous year on campus. Many student groups had to cancel events and publication releases. We are fortunate to have avoided such issues, mostly thanks to advice and support from our faculty advisor, Dr. Jill Shapiro, and the Department of Ecology, Evolution, and Environmental Biology's Kyle Bukhari. Aside from publishing, one of our annual goals in *Sapient* is to share enthusiasm for biological anthropology in the Columbia community. This year, we successfully hosted three events for all students to join us and share their creativity using the medium of human evolution. We would like to extend gratitude to all who attended, as well as an invitation to many more such occasions next year!

Lily will be returning as Co-Editor-in-Chief next year, and Ana will be watching from afar. We are both excited to see what *Sapient* board members and contributors will bring to the table next year and beyond.

Ana Reif and Lily Kuhn, Co-Editors in Chief

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Sapient would like to extend our deep thanks to the Department of Ecology, Evolution, and Environmental Biology at Columbia University for making what we do possible. We are especially grateful for the support of Professor Maria Uriarte, Department Chair; Kyle Bukhari, Director of Academic Administration and Finance; and Professor Jill Shapiro, our faculty advisor and Director of Undergraduate Studies for the Evolutionary Biology of the Human Species program.



Artwork by Ray Atlas

Having a Whiskey with Jane Goodall By Kiley Malloy

She says it's good for her throat, and you'd call bullshit
except she's eighty-nine, you're freshly twenty-two.
Who are you to say anything divisive?

Maybe it's winter, or early spring and the trees speak
through clashing branches like wooden swords. *The leaves
don't fall in the Gombe, you know*, she says and you are made

of longing. Want to slip inside her skin and live there,
just for a while, just to know what it is to feel held
by the roots under your feet. You drink fast,

like your life depends on it while she sips like she has more time
than you. Like every moment is a waste if not spent slowly,
intentionally. Once, an interviewer asked her favorite brand

as if there weren't more important, life-altering things to speak
of. Always the good sport, she said Johnnie Walker Black,
and secretly, you are pleased to have discovered this. You want

to show what you know as if this were a test. You want
to circle the right letter choice, watch her tan jacket
fall from wrist to elbow as she sips away. That forearm,

you think, holds more knowledge than your whole body.
You can't look away, you beg for that type of knowing,
the kind that seeps into your bones. She looks like a person

who knows secrets about the world and maybe about you, too.
You, whose fingertips feel the same as every chimp she's loved. Looks
like she knows what notes to look for in a scotch. You want to ask,

you have the words planned out but instead you say
What about Flint? Does it get easier? You're thinking
of beings who die of a heaviness you've never touched.

And it's here where she abandons words
for something else. Sips, swallows, closes her eyes.
The clearest, most peaceful *No* you've ever seen.

Kin Selection: Necessary Update to Darwinian Evolution or Redundant Hypothesis?

Natalie Sandoval, Columbia University

A Brief Primer Regarding Altruism, Cooperation, and Natural Selection

One puzzle troubled Charles Darwin as he considered his theory of evolution. Evolution was brilliant in its intuitiveness, and natural selection was the key mechanism in this process. Yet Darwin writes of a special difficulty, which, should it remain unexplained within the context of natural selection, “would have proved fatal to my theory” (p. 237). This difficulty stemmed from a recurring observation that seemingly “broke” from the rules governing natural selection: altruism. Altruism, in the biological context, is defined as any behavior that decreases the fitness of the actor and increases the fitness of the recipient (Silk, 2002). If a behavior decreases the actor’s own fitness, and thus their likelihood of reproduction, how could it possibly be preserved through the mechanism of natural selection? Darwin (1859) considered the problem of sterile, eusocial insects which are physiologically distinct from fertile males and females of their species. Again, he ponders how the existence of sterile individuals could fit within the theory of natural selection. Darwin (1859) claimed that this difficulty was negated

by a consideration of the individual as a part of a group, stating that natural selection could be applied to both the individual and the family unit, reconciling the seeming contradictions of altruism. Here, altruism is reduced to its purest form, as the individual is totally biologically prohibited from direct reproduction and expends their energy over a lifetime by aiding others in their group. In the *Descent of Man*, Darwin (1871) would further clarify his understanding of how altruistic traits evolve: “Natural selection sometimes acts indirectly on the individual, through the preservation of variations which are beneficial only to the community” (p. 155). Operating without a modern understanding of genetics, Darwin proposes a theory of altruism almost akin to group selection. Darwin’s intuition was incorrect. With 20th century innovations in genetic science, a new, controversial theory of altruism (and cooperation) gained prominence: kin selection.

J.B.S. Haldane summarized the logic of kin selection in a pithy line: “I would gladly give up my life for two brothers or eight cousins.” He suggests that the size of the incentive to help another is proportional to the degree

of relatedness between actors– a sentiment formalized in mathematical terms by W.D. Hamilton in 1964. Hamilton’s Rule (1964) predicts that genes should increase in frequency when Relatedness (r) multiplied by Benefit to the Recipient (B) is greater than Cost to the Altruist (C), as seen below:

$$rB>C$$

Hamilton’s (1964) explanation of altruism improves upon Darwin’s explanation of the phenomenon by incorporating gene theory: natural selection privileges *genetic* success, the hypothesis proposes, not necessarily reproductive success. This contention is known as inclusive fitness theory (Chapais and Berman, 2004) The basic prediction of kin selection theory is that social behavior should correlate with genetic relatedness and that, in particular, altruistic actions should be more likely to be directed towards relatives. When evaluating the strength of kin selection theory, one must take care to differentiate cooperative and altruistic behavior. Chapais (2006) notes that unilateral altruism is a powerful test of kin selection, because the actor is (definitionally) not receiving any direct benefit from their action. From this, he argues that the impact of kinship on cooperation should be much less extensive than its impact on altruism. Cooperation between individuals may be

influenced by myriad factors, of which Chapais (2006) identifies competence as highly significant in certain contexts.

This paper will explore the predictive value of kin selection theory by evaluating key studies within the scientific literature on altruistic and cooperative behavior in primates. To this end, this review considers both experimental and observational studies, and will outline the merits and limitations of both methods of research. The paper begins with a brief discussion of the principle mechanisms underlying kin selection– the greenbeard effect, kin discrimination, and population viscosity– as these mechanisms are assumed priors in a multitude of studies. The review then turns to cooperation, examining kin selection in the context of coalitions and considering the confounding effects of parental bias, competence dependent decisions, and over generalizations of primate behavior. There are many proposed instances of cooperative behavior among primates, including but not limited to coalition formation, social partner choice, and hunting. Likewise, behaviors such as grooming have been forwarded as examples of unilateral altruism. Yet identifying purely altruistic acts has proved a difficult task and perhaps one best suited to a laboratory environment. This review goes on to respond to this difficulty, considering

how to quantify the elements of Hamilton's Rule and evaluating evidence of kin-biased altruism among specific primate populations.

Mechanisms of Kin Selection

Kin selection is thought to operate via three primary mechanisms: the green-beard effect, kin discrimination, and population viscosity. Understanding the mechanisms underlying kin selection offers greater insight into the theory itself, as well as the experimental studies designed to isolate factors related to altruism and cooperation (Kay *et al.* 2019).

Hamilton (1964) postulated that a cluster of genes might be responsible for selective altruism. He considered a thought experiment wherein members of a group share a gene that encodes for two phenotypes: the first, one which results in a perceptible phenotypic marker; the second, one which causes its carrier to behave altruistically towards those with the phenotypic marker. If individuals are able to recognize shared alleles via phenotypic patterns, and extend preferential treatment towards other allele carriers, it would facilitate altruism between allele-sharing individuals. Dawkins (1976) later termed this process the "Green Beard Altruism Effect," describing a scenario in which individuals with green beards extended preferential

treatment towards others with green beards.

Kin discrimination, sometimes referred to as kin detection, is an organism's ability to differentiate close genetic kin and non-kin (Penn and Fromen, 2010). This ability also plays a central role in inbreeding avoidance; Kin discrimination as a means of facilitating nepotism may simply be an additional benefit. This differentiation may occur by a few different means. Close association during critical periods in early life may induce familiarity between individuals which facilitates discrimination between kin and nonkin (Silk, 2002). Visual and olfactory cues may be important as well, because phenotypic and genetic similarity are highly correlated (Smith, 2014). In a study of ring tail lemurs, Boulet (2009) found that signals of genetic relatedness materialized via odor profiles during the competitive breeding season. These odor profiles allowed related individuals to avoid inbreeding with one another.

Evidence from studies examining familiarity in primates in experimental settings supports the contention that kin detection depends on familiarity established in early life (Silk, 2002). Erhart *et al.* (1997) removed baboons from their mothers as infants and reintroduced half-siblings to one another at ninety days of age. They also introduced the baboons to non-kin

peers. The authors found that kin did not have significantly more affiliative, sociosexual, or agonistic interactions with one another than they did with nonkin, concluding that baboons do not recognize kin without maternal associations in infancy. Likewise, Welker *et al.* (1987) attempted to distinguish behavior stemming from familiarity versus relatedness among macaques. The authors contested claims made by Wu *et al.* (1980) that association is not prerequisite for kin recognition by designing an experimental study of male and female monkeys raised homogeneously. The authors separated infants from their mothers immediately after birth. At five weeks of life, subjects were placed into two separate groups. When the subjects were between 13-42 months old, the authors merged the two groups, creating an environment where the monkeys could interact dyadically with four types of individuals: familiar/nonkin, familiar/kin, nonfamiliar/nonkin, and nonfamiliar/kin (Welker *et al.*, 1987). They found that, above all, individuals preferred to interact with familiar individuals. There was no difference in the rate of interactions between familiar/kin and familiar/nonkin. The authors concluded that familiarity, not kinship, was the determining factor in social partner choice.

Researchers have also considered population viscosity as facilitating kin selection.

Population viscosity describes the occurrence of individuals remaining near their place of birth in maturity, or a limited dispersal (Rodrigues, 2016). This limited or nonexistent dispersal results in a population with a high degree of genetic relatedness between any two given individuals, increasing the benefits of altruism between neighbors. Hamilton (1964) does note that kin competition emerges as a greater likelihood among viscous populations, but maintained this effect would not be prominent enough to totally undermine kin selection. Platt and Bever (2009) have argued that ecological factors such as local density, global density, and local carrying capacity constrain the degree to which kin competition reduces the benefits of cooperation. Under different circumstances, kin bias may be more or less adaptive. This finding is applicable to discussions of kin selection in general, which may wax and wane in importance given the context of an individual's life history, his species, and the precise nature of the cooperative/altruistic behavior.

Cooperation: Coalitions, Competence, and Parental Bias

Though cooperation between primates may net benefits to both individuals, kin selection theory predicts that, all other factors held equal, individuals should favor their kin as

cooperative partners. Any benefit derived from a cooperative activity, such as group hunting or alliance formation, should be magnified if the cooperative partners are related to one another. However, holding all other factors equal has proven challenging. Drawing conclusions from observations of primates interacting in their native habitats involves considering a storm of confounding factors. Though kinship appears to be one element that influences how a primate picks his cooperative partner, it may, in the right contexts, be overshadowed by other considerations (Chapais, 2006).

Take, for instance, coalition formation. A coalition, or an alliance, can be defined as a coordinated attack by two or more individuals against one or more opponents (Chapais, 1995). Coalitions are an important tool in evaluating kin selection theory as they are an obvious example of individuals benefiting others at a cost to themselves (even if they risk relatively minor injuries) (Bernstein, 1991). Intervening in a fight, or agonistic aiding, is costly to the actor as they spend (potentially significant) time and energy in this pursuit rather than other activities and put themselves at risk of physical harm (Smith *et al.*, 2010). Given this cost/benefit structure, Hamilton's Rule predicts that individuals will intervene more often for kin than nonkin. This

prediction is consistent with studies that suggest kin generally bias coalitionary support towards genetic relatives. For instance, in Thierry *et al.*, (2004), Chapais writes that, in macaques, the formations of coalitions during conflict is more frequent among matrilineal kin. Kinship seems to play an especially important role when confronting a high-ranking aggressor. Not all coalitions pose the same risk to the intervening actor and intervening in an agonistic interaction against a high-ranking opponent may pose a greater social and physical risk to the coalition partner. In an laboratory study of Japanese macaques, Chapais *et al.*, (1997) found that individuals were more likely to aid their close kin against higher-ranking opponents than they were to intervene on behalf of distant kin against higher-ranking opponents. The authors observed that grandmothers intervened at a much greater rate on behalf of their grandchildren than did aunts on behalf of their nieces, indicating that the difference in the nature of the kinship between individuals does translate into differential rates of nepotistic aiding (Chapais *et al.*, 1997). Further evidence comes from Bernstein and Ehardt, (1985), who analyzed over 3,500 episodes of agonistic aiding in rhesus monkeys over a two year period. The authors found that females defended or aggressively supported kin versus

third parties without regard to their hierarchy. From this, they concluded that most defensive aiding was for the purpose of defending victims, who were mostly kin, from aggression. Bernstein and Ehardt, (1985) note that mothers aided their children significantly more often than any other class of relationship, emphasizing the necessity of differentiating kin bias from parental bias. It is clear, at the very least, that mothers often behave preferentially towards their offspring, even when offering aid comes at some risk or cost to themselves.

One might reasonably question how many instances of observed "kinship bias" really represent instances of parental bias. Even Darwin understood that mothers act preferentially towards their offspring via parental care. The innovation of kin selection was to, using gene theory, offer an explanation for why extended kin would behave in nepotistic ways. In order to evaluate the prevalence of parental bias among all cases of kin bias, one must distinguish parental interactions (usually mothers and offspring) from other kin interactions. Bernstein and Ehardt, (1985) provide this level of analysis, but this detail is lacking from certain papers reviewing the literature on kin bias; For instance, Smith (2014) refers to a kin bias across the primate order with regards to coalitionary support. She

notes that 94% of primate species offer this support to individuals they are most related to, without specifying if this includes parents and children. In one revealing study, Ehardt, (1988) found that female sooty mangabeys groom, were approached by, were in proximity to, and were in contact with their kin significantly more than would be predicted by random chance-- but all significant kin effects were eliminated once mother-infant interactions were excluded from the data. Ehardt, (1988) notes that consideration of parental bias has been neglected by other authors, though some researchers reported difficulty in distinguishing preferential affiliation among distant kin.

Chapais, (2006) claims that cooperative activities can be further distinguished into two categories: low-competence activities and competence-dependent activities. Where the latter is concerned, Chapais, (2006) argues that individual primates choose their cooperative partner primarily on the basis of competence, not kinship. He offers the example of an individual who wants to gain access to resources which are currently being hoarded by a high-status individual. The lower-status individual offers the higher-status individual grooming as a means of attaining tolerance. In this hypothetical, the status of the individual is the deciding

qualification with regards to grooming, not degree of relatedness. This interpretation of the cooperative act aligns with what Cheney (2011) terms “short-term byproduct mutualism,” in which an individual simply assesses the present value of a potential partner before deciding whether to engage in trading (p. 10904). Cheney (2011) notes that, though some social interactions do change according to the current value of a partner, much social interaction between primates occurs in the context of long-term affiliative relationships. In this context, short-term asymmetries are acceptable.

Chapais (2006) goes on to group reciprocal grooming, allomothering, and huddling together as low-competence activities. He distinguishes competence dependent activities as grooming for tolerance, grooming for aiding, mutual aiding, mutual sharing, social play, and group hunting. Here, as in other papers, Chapais (2006) does not totally discount the theory of kin selection. He merely suggests that its power has been overstated and that kin bias is one factor among many in driving primate behavior. Other elements that determine the distribution of kin bias are, Chapais and coauthor Bélisle argue, the cost/benefit ratio of an action, the number of kin in the group, and the degree of dependence individuals have on their social partners (Chapais

and Bélisle, 2004). Because of the diversity of social structures across the primate order, one might reasonably infer that distribution of kin bias will likewise vary, and may play a greater role among some species than others.

Indeed, as Strier (1994) claims, the myth of the “typical primate” has confused much of the discussion of kinship in primates (p.233). When one discusses ‘primate behavior’, one actually refers to a broad swath of behaviors, which, when taken collectively, lead one to contradictory and unhelpful conclusions. Strier (1994) offers the example of chimpanzees and baboons as two species with significant differences in social organization and behavior. Baboon females typically have more stable intra-sexual relationships than baboon males and affiliative interactions, including matrilineal kin coalitions, are essential (Strier, 1994). On the contrary, chimpanzee males form stronger intra-sexual bonds than do chimpanzee females, likely a result of female dispersal (Strier, 1994). These differences are extraordinarily important in any discussion of kin bias, which may depend on some form of kin recognition established via close association early in life (as discussed in “Mechanisms of Kin Selection”).

Studies, such as the aforementioned Erhart *et al.* (1997) and Welker *et al.* (1987),

cemented the importance of familiarity in distinguishing kin from nonkin in primate species. These findings, in conjunction with the tendency for males to disperse from their natal group in many primate species, resulted in a widespread belief that primates would not recognize paternal kin. Female kinship was viewed as the core organizing principle to consider when examining social behavior in primates (Gouzoules, 1984). Contrary to this view, recent studies provide evidence that kin recognition (and bias) may occur among patrilineal kin. For instance, Parr and de Waal (1999) discovered that chimpanzees females can match photographs of unfamiliar females with their male offspring, but not with their female offspring. The authors interpreted this finding as an adaptive response to the patrilineal structure of chimpanzee societies. In their study of rhesus macaques on Cayo Santiago, Widdig (2001) observed maternal half-sisters, paternal half-sisters, and unrelated females, comparing the rate and type of social interactions between individuals. As predicted, the maternal half-sisters favored one another as social partners, spending a greater proportion of time grooming them and in close spatial proximity to them as compared to the unrelated females. Unexpectedly, these rhesus macaques also spent more time grooming and in close

proximity to their paternal-half sisters. From these results, Widdig (2001) concluded there is positive evidence for the claim that macaques can recognize paternal kin. Potential mechanisms include phenotype matching, the influence of dominance rank on affiliation patterns, and social bonding between half siblings encouraged by mothers in childhood (Widdig, 2001).

Altruism: Finding Unilateral Altruism and Controlling for Confounds

Unilateral altruism may be the finest test of kin selection, but identifying altruistic behavior, and its frequency in nature, have proven difficult. Proponents of kin selection theory, such as Foster *et al.* (2005), claim that true altruism can evolve only under conditions of relatedness. Even Darwin himself seems to have agreed with this sentiment. He writes, in *On the Origin of Species*, that natural selection absolutely cannot produce adaptations in a species that exclusively benefit another species (Darwin, 1859). Another glance at Hamilton’s Rule supports this claim- relatedness must be greater than zero in order for the inequality to function (Silk, 2002). Foster *et al.*, (2005) go on to argue that all attempts to distinguish altruism between non-related individuals have been so flawed as to be rendered worthy of dismissal. For instance, they critique studies which draw limited definitions

of “kin,” excluding common ancestry that extend beyond immediate siblings (Foster *et al.*, 2005). Foster *et al.* (2005) are specifically responding to biologist E.O. Wilson, once a firm proponent of kin selection among insects, who later argued that a “eusocial gene” was a better theory for the source of altruistic behavior than kin selection. It bears mentioning that Wilson’s research focused on eusociality in insects, a decidedly different order than primates. His critiques do not transfer cleanly onto primates, which typically do not have obligatorily sterile group members. Silk (2002) notes that unilateral altruism is distinct from reciprocal altruism, a relationship between individuals in which altruistic acts balance over time. Reciprocal altruism works via collaboration for mutual benefit over the long term, and therefore does not require the altruistic partners to be kin (Silk, 2002). Return to, for instance, the green-beard effect, which would facilitate altruism among individuals with a shared phenotypic trait.

Other difficulties emerge when attempting to quantify the individual components of Hamilton’s Rule (cost to the altruist, benefit to the recipient, and degree of relatedness). Behaviors which were once considered classic examples of unilateral altruism, such as grooming, have come under heavy criticism, with studies such as

Shutt *et al.* (2007) suggesting both cooperative partners derive some benefit from the act. Shutt *et al.* (2007) performed an observational study of grooming in barbary macaques, finding novel evidence that grooming is correlated with lower levels of stress in the *provider* of the grooming. These conclusions contradict a once prominent strain of thinking which contended that grooming was altruistic because it represented an opportunity cost in terms of time spent foraging. Kurland (1977) notes that grooming may have various functions between individuals depending on the context of the interaction, including coalition formation, stress reduction, and social manipulation.

Then there is the problem of assessing the impact of a behavior on an individual’s lifetime fitness (Silk, 2002). How can one quantify the benefit to the recipient and the cost to the altruist over a life history that spans decades? Grafen (1991) provides an imperfect but useful solution in the “phenotypic gambit,” a term in evolutionary game theory which proposes that the short term benefits from altruistic acts become long term distinctions in fitness between individuals. Even the degree of relatedness between individuals, which appears far more quantifiable at first glance, has proved tricky to assess in practice. Due to the tendency towards

male dispersal, researchers have been historically limited to reconstructing matrilineal lines, a strenuous process of tracing genealogies based on observations of spatial associations and nursing (Silk, 2002). New tools for determining genetic relationships in primate populations make assessments of relatedness more accurate, and expand the possibilities for patrilineal kinship studies. Pedigree construction involves evaluating the cooperation of individuals with paternal and maternal offspring and descendants, as well as more distant kin (Smith, 2014). Tracking microsatellites, pieces of DNA that include recurring motifs, can also be used to determine degrees of relatedness between individuals and populations (Smith, 2014).

One confounder which impedes conclusions drawn from observational studies of primates is proximity. Authors such as Smith (2014) cite social partner choice as evidence of kin bias. Yet it may be that primates simply tend to have closer spatial associations with their kin as compared to non-kin individuals, making kin a more convenient choice of social and/or cooperative partner. If primates tend to prefer whatever individuals are closer to them, it would not provide strong evidence for kin selection. Griffin and West (2002) emphasize that high relatedness between individuals does not suffice

as evidence for kin selection. The authors note that there may be a high degree of intra-group relatedness– but in species with limited dispersal, relatives will compete for resources with one another, reducing the overall significance of inclusive fitness.

Kurland’s (1977) study of female Japanese macaques addresses the critique of kin selection from the perspective of proximity. Kurland (1977) observed that female monkeys spent more time grooming and in proximity with kin than nonkin, but also had more agonistic interactions with kin than nonkin. He does suggest that the intensity of the agonism directed towards kin and nonkin may vary. Silk *et al.* (1981) reported that rates of aggression among maternal kin in primates were as high, or higher, as compared to rates of aggression among those of different matriline. When they distinguished between types of agonistic behavior– threats, displacements, chases, and attacks– the authors found that the most severe types of aggression were generally directed towards nonkin (Silk *et al.*, 1981). In a review that considered mammals and primates of all species, Smith (2014) found that most primates exhibit higher rates of aggression towards their kin. Were primates picking their grooming and social partners on the basis of kinship, as kin selection theory would predict, it follows that they

should also avoid picking agonistic interactions with kin, as fighting with kin confers a fitness loss to both individuals. These findings seem to offer evidence against kin selection theory. However, Kurland (1977) went on to control the rate of agonistic interaction for proximity. The researchers found that the frequency of agonistic interactions between kin was proportionately less frequent than the frequency of agonistic interactions in the general population of female Japanese macaques (Strier, 1994).

Experimental studies offer the opportunity to control for factors like proximity and give individuals the opportunity to engage in purely altruistic behaviors, often a rarity in natural environments. Silk (2006) identifies food sharing as one clear example of unilateral altruism. Schaub (1996) tested altruism among long-tailed macaques via a food-sharing experiment, giving female macaques the opportunity to capitalize on a food source for themselves or share with a partner. He constructed a testing apparatus that allowed a potential donor to manipulate a bar in order to give a partner access to popcorn. Kin selection theory predicts that, should an individual share their food at all, they would behave more altruistically towards kin partners than towards nonkin partners. Schaub (1996) interpreted “greater altruism” to mean sharing food sources

longer with kin than nonkin. He found no significant pattern of preferential treatment of kin over nonkin. Only three of the eleven donors gave their kin access to the food source for a statistically significant time over nonkin. Of the three donors who did demonstrate a kin bias in food sharing, Schaub (1996) notes that they allowed their offspring to have more food as compared to other juvenile nonkin. This finding, again, points to the significance of parental bias rather than kin bias. Schaub (1996) found no evidence for “Hamiltonian altruism,” meaning, altruism between non parent/offspring dyads. Silk (2002) critiques Schaub’s (1996) methodological approach of a laboratory experiment, noting female long tailed macaques do not regularly demonstrate food-sharing behavior. Bélisle and Chapais (2001) also conducted a food-sharing experiment in a laboratory setting with macaques. They found some positive evidence for kin selection, observing that the frequency of food sharing increased as degree of relatedness between individuals increased (Bélisle and Chapais, 2001). Notably, females did not share with their aunts or nieces, just their daughters, grandchildren and sisters. Altruism among extended kin appears a far rarer phenomenon.

Conclusion

One cannot consider primate behavior in the absence of cooperative behavior. Primates, across the order, are social creatures, organizing their societies in a diversity of structures which demand interactions between individuals. They form bonds which often appear remarkably similar to the relationships between mothers and children, family, and friends, in human beings. Thus the origins of cooperation in primates, including *Homo sapiens*, is a question deserving of considerable investigation. Cooperation is sometimes mistaken for its decidedly more enigmatic cousin, altruism. Altruism, especially of the unilateral variety, is distinct from cooperation. It is a puzzle which provokes the question: how could nature select for a behavior which, by its very definition, reduces the fitness of its carriers? Kin selection emerged as a unified theory which promised to account for both cooperation and altruism under the logic of inclusive fitness. The principles of Hamilton’s Rule certainly ring true across time. Even still, experimental and observational studies yield evidence that speaks to the limited applicability of kin selection, and the relatively weak strength of its predictive powers.

Kinship may play a role in determining coalition formation. Yet Bernstein and Ehardt’s (1985) finding of kin-support among female macaques is somewhat complicated by further analysis which found that the most common form of support was mothers intervening on behalf of their children, an example of parental, rather than kin, bias. Ehardt (1988) provides further support for this contention. There is perhaps a greater conflation of parental bias with kin bias in the literature which has yielded unfounded support for kin selection. Still, kinship is hardly the only factor that determines a primate’s choice of cooperative partner. Chapais (2006) notes the importance of distinguishing between low-competence activities and competence-dependent activities, which have different cost/benefit structures. Degree of relatedness is simply one factor that influences partner choice and is of flexible importance depending on the nature of the given behavior. This admission makes interpreting observational studies even more difficult, as there are many confounds which might determine any particular behavior.

Strier (1994) notes the importance of distinguishing between behavior in various species of primates, moving away from anthropological biases which have historically dominated primatology. Kin selection may play a more

prominent role in certain species, depending on their dispersal and social living patterns. Kin competition can vastly reduce the benefits of cooperating with one's kin. The strength of kin selection may vary not only by species, but by kind of behavior. A basic distinction between altruism and cooperation is necessary. Within altruistic behavior, there exists unilateral altruism and reciprocal altruism, which may have different functions and mechanisms. Perhaps kin selection remains the strongest explanation for unilateral altruism– but one must question how often unilateral altruism occurs in nature, and how often it occurs between kin that are not mother/child dyads. Experimental studies such as that conducted by Schaub (1996) lend credence to the rarity of unilateral altruism among primates.

Despite emerging evidence that detracts from the explanatory power of kin selection, the biological and cultural centrality of kinship in organizing human societies underscore kinship as a pertinent area of inquiry (Strier, 1994). The influence of nepotism and relatedness on primate behavior remains greatly relevant in assessments of primate social structure. Given the significance of kin in *Homo sapiens*, it is not unreasonable to look to other primates for clues as to the origin of these relationships.

Appendix

When assessing the role of kin selection in provoking cooperative and altruistic behavior in primates, one must engage with cognitive models. The degree to which kin recognize each other based on intuitive triggers– such as smell– as opposed to advanced mental models of individuals (theory of mind) is not totally certain. Likewise, one must consider the intricacies of primate cognition when examining the question of friendship among non-human primates. The two inquiries seek to explain the nature of affiliative, long term relationships between individuals, and what mechanisms might underlie them. Friendship and family are pillars of the human experience, making it an intriguing avenue of study in other species of the primate order. Some parallels emerge between examples of reciprocal altruism and examples of friendship among primates. Questions of short and long term benefit to both individuals emerge. Asymmetries tend to balance over time, else the relationship collapses. Might “friendship” simply be a relationship with an expectation of reciprocal altruism, drawn out over some extended period of time? It is certainly evident that prolonged exposure to other individuals, especially in infancy, can create powerful bonds between non-kin primates that

lead to cooperation and even altruism.

Turning to methodological considerations, there are many applications of the study of primate cognition to the study of kin selection, bias, and recognition. In each case, one must consider the confounds that accompany the chosen method of study. Observational field study, experimental captive study, and experimental field study all yield unique benefits and challenges and may be more appropriate for addressing certain questions than others. In the case of kin selection, experimental captive studies offer the opportunity to induce opportunities for altruism which may be so rare in the field as to be statistically insignificant.

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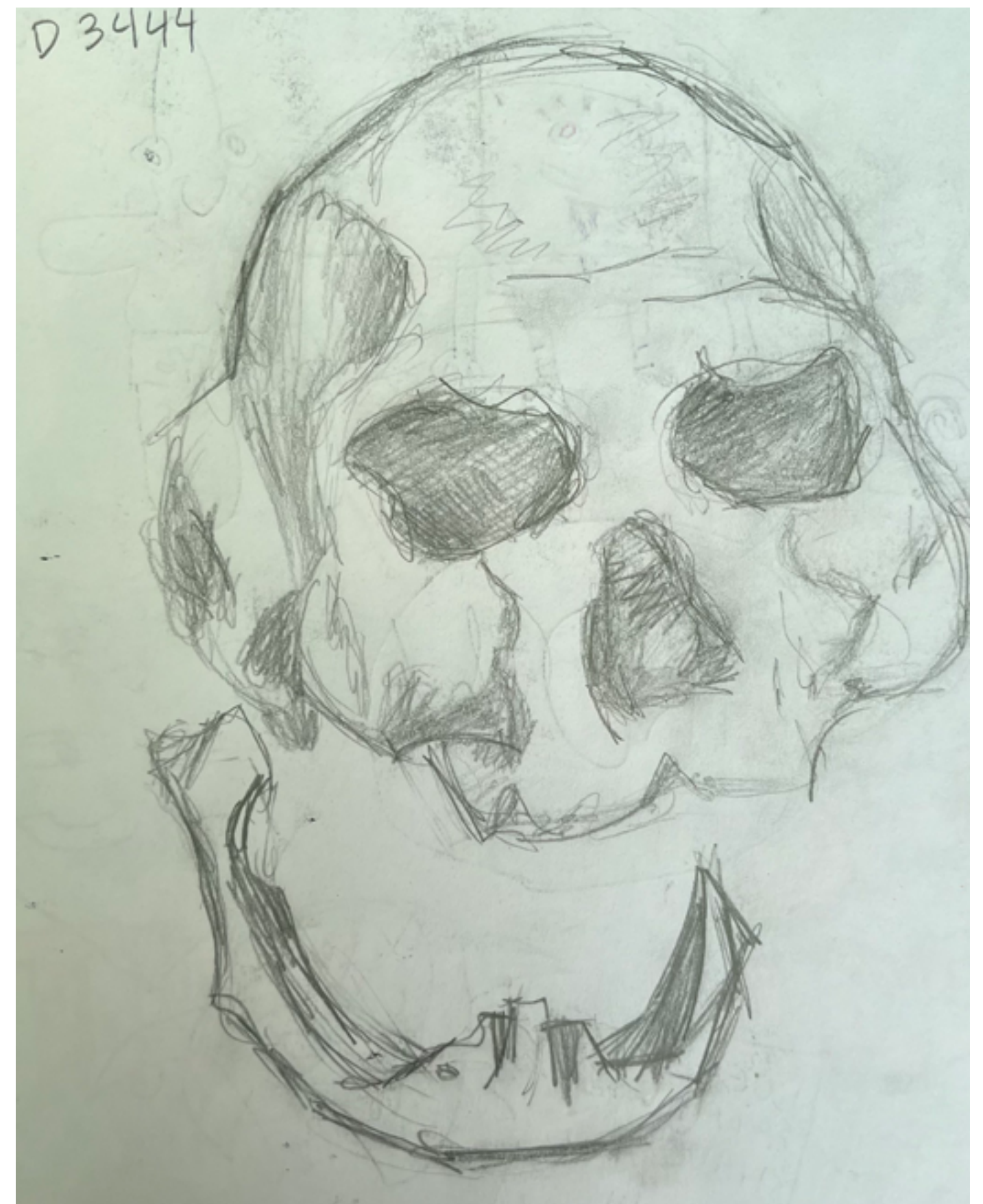
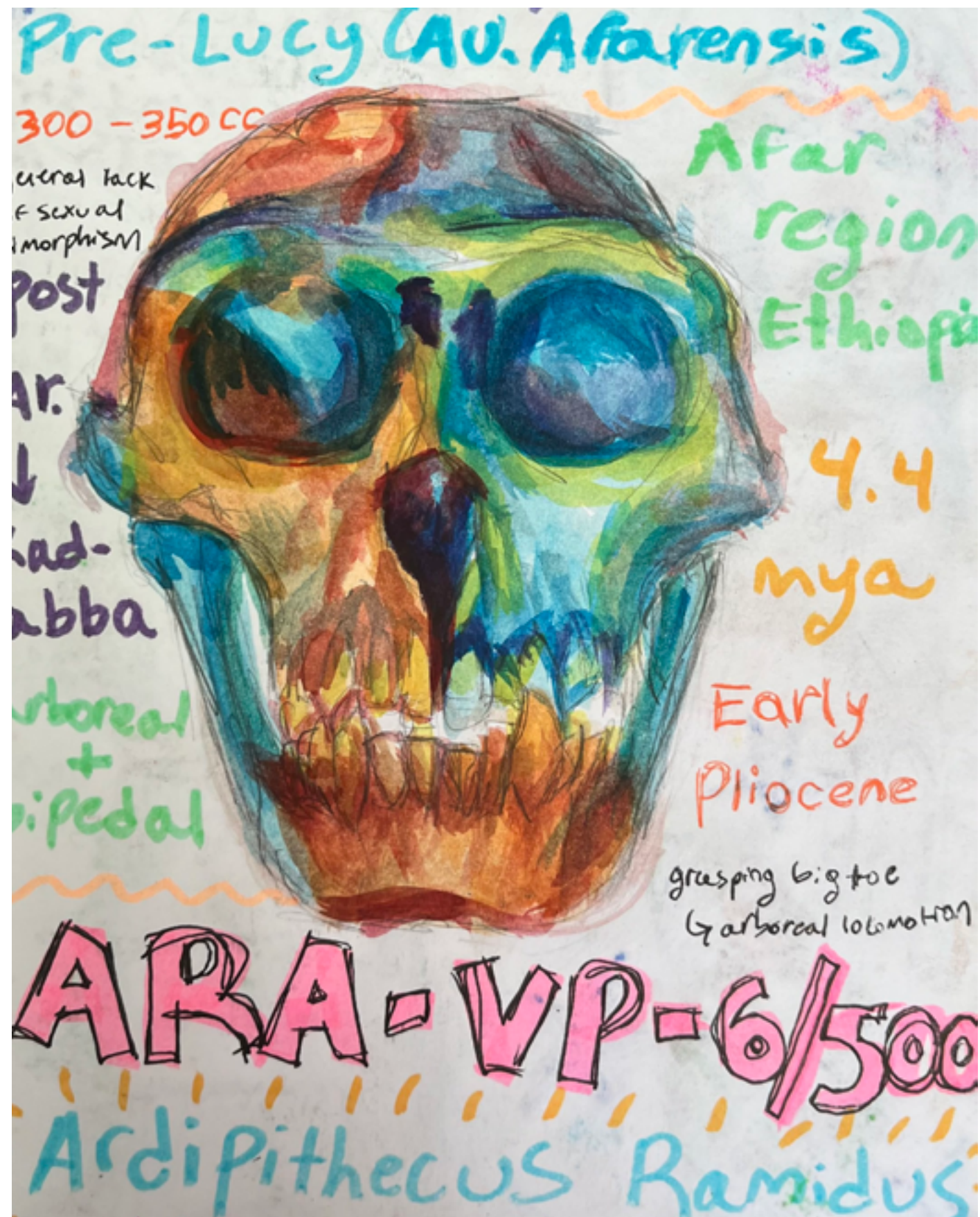
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Artwork by Ray Atlas

Linear Enamel Hypoplasia in Nonhuman Primates

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Introduction

Researchers disagree on the causes of linear enamel hypoplasia among nonhuman primates (cited in Guatelli-Steinberg, 1998). It is clear that consistent physiological stress related to nutritional deficiencies is a likely cause of many hypoplasias, but these physiological stressors are themselves varied, ranging from environmental shifts (Chollet and Teafor, 2010; Dirks *et al.*, 2002; Guatelli-Steinberg and Skinner, 2000; Tsukamoto, 2004) to disease (Chollet and Teafor, 2010; Skinner and Hopwood, 2004) to physical development (Dirks *et al.*, 2002). Add in genetic explanations, enamel wear with age, and differences in expression across and within taxa, and it becomes apparent that the causes of linear enamel hypoplasia are different for *each individual*. This paper will discuss linear enamel hypoplasia beginning with the formation of healthy enamel, and explore hypotheses for the causes of the defect and its variation throughout the Primate order.

Healthy Enamel and the Formation of Defects

Dental enamel is the very hard, mostly inorganic tissue that covers the crown (Goodman and Rose, 1990) of primate teeth. It is present on both deciduous and permanent teeth (Goodman and Rose, 1990) and aids in mastication by forming a strong, hardened coating (Durruty and Pilar, 2002). As noted by Goodman and Rose (1990), dental enamel is formed by ameloblasts in a process consisting of two broad steps: amelogenesis and maturation. During amelogenesis, secretory ameloblasts form the structure of the enamel. As the enamel matures, the ameloblasts change shape, leaving behind enamel that is almost entirely calcified and inorganic (Goodman and Rose, 1990). Amelogenesis begins at the “occlusal (incisal, cuspal) tips of the tooth crowns” (Goodman and Rose, 1990, p.62) and forms linearly down the tooth, leaving behind distinct transverse bands (perikymata) that correspond with periods of matrix excretion (Durruty and Pilar, 2002).

The linear growth of enamel allows for the formation of defects as the tooth

grows (Goodman and Rose, 1990). If, during amelogenesis, the ameloblasts are subjected to a prolonged disturbance, developmental dental enamel defects may form (Durruty and Pilar, 2002). Dental enamel does not reshape once it is formed; therefore, any defects that form during enamel growth remain throughout the life of the individual (Durruty and Pillar, 2002). One category of developmental enamel defects, characterized by an abnormal thinning of the dental enamel, is called hypoplasia (Goodman and Rose, 1990) from the Greek *hypo* (under) and *plasis* (growth). These defects vary in appearance from small pits to large grooves of missing enamel (Goodman and Rose, 1990). Linear enamel hypoplasias – thin lines that extend around the tooth and run parallel to, but deeper than, the perikymata (Benderlioglu and Guatelli-Steinberg, 2019) – are the most common type of hypoplasia that occurs as a result of stress (Chollet and Teafor, 2010). The presence and location of linear enamel hypoplasias can indicate the age of the individual and the conditions within which the defect formed (Durruty and Pilar, 2002). Enamel is formed in “short-period increments” of daily growth, allowing for estimations of exactly when a stressful event occurred and when it ended, accurate to the day (Dirks *et al.*, 2002).

Thus, the study of enamel growth patterns and linear enamel hypoplasia can provide detailed information about the lives of the individuals who developed the defect, and the physiological stressors they experienced.

Causes of Linear Enamel Hypoplasia Nutrition

One physiological stressor that is often associated with the development of linear enamel hypoplasia is the lack of sustained access to nutrition (Chollet and Teafor, 2010; Guatelli-Steinberg and Benderlioglu, 2006). This can be the result of the environment, as well as seasonal changes in temperature or rainfall (Chollet and Teafor, 2010). In a study on *Cebus*, conducted by Chollet and Teafor (2010), the percentage of individuals living in tropical rainforest and coastal environments that had linear enamel hypoplasia was significantly lower than that of individuals in a seasonal, semi-deciduous forest (Chollet and Teafor, 2010). In this study, Chollet and Teafor (2010) also found that individuals from warmer climates, with a mean monthly temperature greater than 18 degrees Celsius, had significantly lower levels of hypoplasia than those from more temperate climates, with mean monthly temperatures between 15 and 19 degrees Celsius. These warmer temperatures and

wetter environments may contribute to greater resource availability and more consistent access to nutrition, and thus less physiological stress over time and fewer instances of linear enamel hypoplasia. Furthermore, a study conducted on *Macaca mulatta* by Guatelli-Steinberg and Benderlioglu (2006) found that there were fewer instances of unilateral linear enamel hypoplasia among “post provisioned” macaques (those with additional food supplied) than among “preprovisioned” macaques (those surviving largely on natural resources). These findings suggest that physiological stress associated with lack of access to food is a contributing factor to the development of hypoplastic defects (Guatelli-Steinberg and Benderlioglu, 2006).

Seasonal Shifts

The relationship between seasonal shifts in rainfall and the physiological stress that can lead to linear enamel hypoplasia is disputed. The formation of hypoplasias has been shown to increase in some individuals during rainy seasons (Chollet and Teaford, 2010; Guatelli-Steinberg and Skinner, 2000; Tsukamoto, 2004). Tsukamoto (2004) found that, in bonobos, linear enamel hypoplasia appeared to be regular, periodic, and aligned with regular rainy seasons, as opposed to episodic instances of food insecurity. This

increase in hypoplasia during rainy seasons may have been due to greater parasite activity during these seasons, or a result of foraging independently in new conditions, without the help of the mother (Dirks *et al.*, 2002), rather than a result of the rainfall itself (Guatelli-Steinberg and Skinner, 2000; Tsukamoto, 2004). Chollet and Teaford (2010) also concede that dental erosion—the result of chewing the coarser plants common in drier environments—may obscure the expected relationship between dry conditions, lack of food, and increased occurrences of linear enamel hypoplasia.

This dispute is further illustrated by the contrasting results of studies concerning the differing rates of linear enamel hypoplasia development between Bornean and Sumatran orangutans (Hannibal and Guatelli-Steinberg, 2005; O’Hara *et al.*, 2023; Skinner, 2014). Skinner (2014) found that there was a significant difference in the number of perikymata between linear enamel hypoplasias in Bornean versus Sumatran orangutans, and attributed this to the differing seasonal environments of the two islands. Hannibal and Guatelli-Steinberg (2005) and O’Hara *et al.* (2023), however, found that there was no significant difference in the number of perikymata between hypoplasias, nor were there differences in the number

of hypoplasias themselves, indicating that seasonal variation may not be the sole cause of the development of hypoplasias among these individuals. Further research is necessary to understand the relationship between seasonal shifts in temperature and precipitation and the development of linear enamel hypoplasia.

Disease

Disease is another physiological stressor that can contribute to a prolonged period of nutritional stress and the development of linear enamel hypoplasia in nonhuman primates (Skinner and Hopwood, 2004; Chollet and Teaford, 2010). Skinner and Hopwood (2004) present malaria and hookworm as possible causes of hypoplasia. Both malaria and hookworm cause symptoms in young apes that could lead to “severe gastrointestinal upset,” leading to the nutritional deficiencies that cause linear enamel hypoplasia (Skinner and Hopwood, 2004, p.230). The disease hypothesis is also disputed, as Skinner and Hopwood (2004) contend that linear enamel hypoplasia is more likely a result of malnutrition than disease. Chollet and Teaford (2010) found that individuals living in warmer, more tropical environments, where the spread of disease may be more prevalent, did not exhibit a higher frequency of linear enamel hypoplasia, a finding

that refutes the disease hypothesis.

Social Stressors

Linear enamel hypoplasia can also develop in response to social stressors, specifically those related to changes during sexual maturation and development (Dirks *et al.*, 2002). In a study conducted by Dirks *et al.*, (2002), female baboons developed linear enamel hypoplasias at similar stages of development. Hypoplasia development corresponded to periods of independence from their mothers, such as weaning, and periods of sexual maturity, such as menarche. Independence from the mother causes nutritional stress and thus raises the likelihood of developing hypoplastic defects, as a result of independent food search and change in diet, conflict with the mother over weaning, and an increase in potential exposure to pathogens (Dirks *et al.*, 2002). In female baboons, linear enamel hypoplasias develop at menarche and in the period following, as a possible result of psychological stress (Dirks *et al.*, 2002).

Other Factors and Confounding Variables

While prolonged physiological and social stress is strongly correlated with the development of linear enamel hypoplasias, other factors may also contribute to the development of these

defects. For example, although such occurrences are rare in modern populations, parents may pass hypoplastic defects to their offspring, and could thus occur regardless of physiological stressors (Goodman and Rose, 1990).

Additionally, many studies (Benderlioglu and Guatelli-Steinberg, 2019; Goodman and Rose, 1990; Teafor and Chollet, 2010) have found that after enamel on permanent teeth fully develops, there is a negative relationship between the presence of linear enamel hypoplasia and age, with younger individuals having more hypoplasias than older individuals; therefore, age can be a significant confounding variable in the analysis of defects. There are two major hypotheses as to why this relationship appears to exist. The first hypothesis posits that older individuals accrue more enamel wear over time, obscuring both the perikymata and any linear enamel hypoplasias that may have formed (Benderlioglu and Guatelli-Steinberg, 2019; Teafor and Chollet, 2010). The second hypothesis suggests that linear enamel hypoplasias are indicative of the severe and prolonged stress that may have resulted in the premature death of the individual (Goodman and Rose, 1990), and that those who do not have linear enamel hypoplasia never experienced the same stress. The negative relationship between

age and hypoplasia frequency after formation demonstrates the need to consider multiple factors when determining the cause of hypoplasia in an individual.

The presence of linear enamel hypoplasia symmetrically across several teeth is often evidence of prolonged stress (Goodman and Rose, 1990). However, hypoplasias can also occur due to localized trauma, which can cause asymmetrical hypoplasias on antimeric teeth (Benderlioglu and Guatelli-Steinberg, 2019). This localized trauma can occur if one side is subjected to more stress than the other, or if one tooth is more effectively able to resist stressors (Benderlioglu and Guatelli-Steinberg, 2019). The presence of asymmetric defects and the possibility of localized trauma indicates that the expression of linear enamel hypoplasia can vary even among individuals who experience the same stressor. Furthermore, as noted by Guatelli-Steinberg and Benderlioglu (2019), the exclusion of asymmetric defects could limit or skew the findings of continued studies on linear enamel hypoplasia.

Variation

There is significant variation in the presence and timing of hypoplastic defects across the Primate Order (Guatelli-Steinberg,

2000; Newell, 2006; Tsukamoto, 2004).

Across taxa, this variation is often the result of differences in life history, dental eruption time, and speed of enamel growth (McGrath *et al.*, 2019; Newell, 2006).

Generally, linear enamel hypoplasia is more prominent in Great Apes than in monkeys (Guatelli-Steinberg, 2000), with bonobos, in particular, being very stressed (Tsukamoto, 2004). One explanation for the increase in linear enamel hypoplasias in great apes is their longer life histories and prolonged dental eruption and development, giving more time for individuals to experience stress and accumulate enamel defects (Guatelli-Steinberg and Skinner, 2000; Newell, 1998). A study conducted by Newell *et al.* (2006) supports the hypothesis that longer life histories contribute to increased hypoplastic defects by demonstrating that even among the *Ceboidea*, the longer-lived *Cebus* had higher frequencies of linear enamel hypoplasia than the shorter-lived *Callicebus*, *Cebuella*, and *Saimiri*. Gibbons are also of particular interest in this case, as they have a life history that falls between that of monkeys and great apes, and demonstrate patterns and frequencies of linear enamel hypoplasia between the two groups (Guatelli-Steinberg, 2000).

Variation *within* taxa may also obscure differences in the frequency of linear enamel hypoplasia between taxa. Among the *Cercopithecoidea*, for example, there is a wide array of enamel thicknesses, with *Pongo* having the thickest enamel and mountain gorillas having the thinnest (Cited in McGrath *et al.*, 2019). As demonstrated by McGrath *et al.* (2019), the relatively thin enamel of mountain gorillas leads to significantly shallower hypoplasias that can be easily worn away or overlooked, leading to them being undercounted. While not all great apes share the thin enamel of the mountain gorilla, this study highlights the degree to which individual variation and variation within taxonomic groups can affect the expression and visibility of hypoplastic defects.

It is difficult to draw conclusions about the emergence and patterns of linear enamel hypoplasias based on taxon alone. At a grade level, while great apes generally have more instances of linear enamel hypoplasia than monkeys, there are enough exceptions and variations within the grades (Guatelli-Steinberg and Skinner, 2000) to render a grade-level analysis insufficient on its own. There is variation among monkeys as well, with more than half of the individuals in a sample of red colobus monkeys developing linear enamel hypoplasia,

while some species of African monkeys developed no hypoplasia at all (Guatelli-Steinberg and Skinner, 2000). Beyond a grade-level classification, there are variations among even closely-related and interbreeding species. Dirks *et al.*, (2002), for example, found no evidence that hypoplastic defects among four baboons were the result of “anything beyond individual variation” (p.246). While species accounts for some of the variation in the development of linear enamel hypoplasia, researchers must also consider the individual environmental conditions, stressors, and life history to gain a full understanding of the defect.

The development of linear enamel hypoplasia also varies by sex across the Primate Order (Newell, 1998; O’Hara *et al.*, 1993; Kralick and McGrath, 2021). Generally, males tend to have greater frequencies of the defect than females, but this is subject to variation by species as well (Newell, 1998). Among Bornean and Sumatran orangutans, for example, female individuals not only have higher frequencies of defects, but also acquire them more often than males (O’Hara *et al.*, 2023). Additionally, there is significant variation among male individuals of the same species; as Kralick and McGrath (2021) established among male orangutans, for example, flanged orangutans have “significantly deeper

LEH defects” (p.625) than their unflanged, same-species counterparts.

Conclusion

The stressors that cause linear enamel hypoplasia are varied, and it is not possible to determine the precise cause of the defect from its physical characteristics alone (Guatelli-Steinberg *et al.*, 2012). Furthermore, while there is variation among species, there is a strong, consistent correlation between the development of linear enamel hypoplasia and prolonged physiological stressors. A nonhuman primate with linear enamel hypoplasia likely experienced severe and prolonged physiological stress – whether that stress was the result of environmental shifts, development, or disease is not clear from the teeth alone. Variation by geography or sex, or across and within taxa further complicates generalization of the causes of the defect across the order. By all means, it is incredibly difficult to determine exactly *what* temporarily interrupted an individual’s process of amelogenesis.

However, it is exactly because of this wide variation that linear enamel hypoplasia provides such highly specific insight into the life of each individual who develops it. By examining

all factors together, the emergence, timing, and patterns of linear enamel hypoplasia can provide insight into the stressors that a long-dead individual experienced, the environment they lived in, and their social organization.

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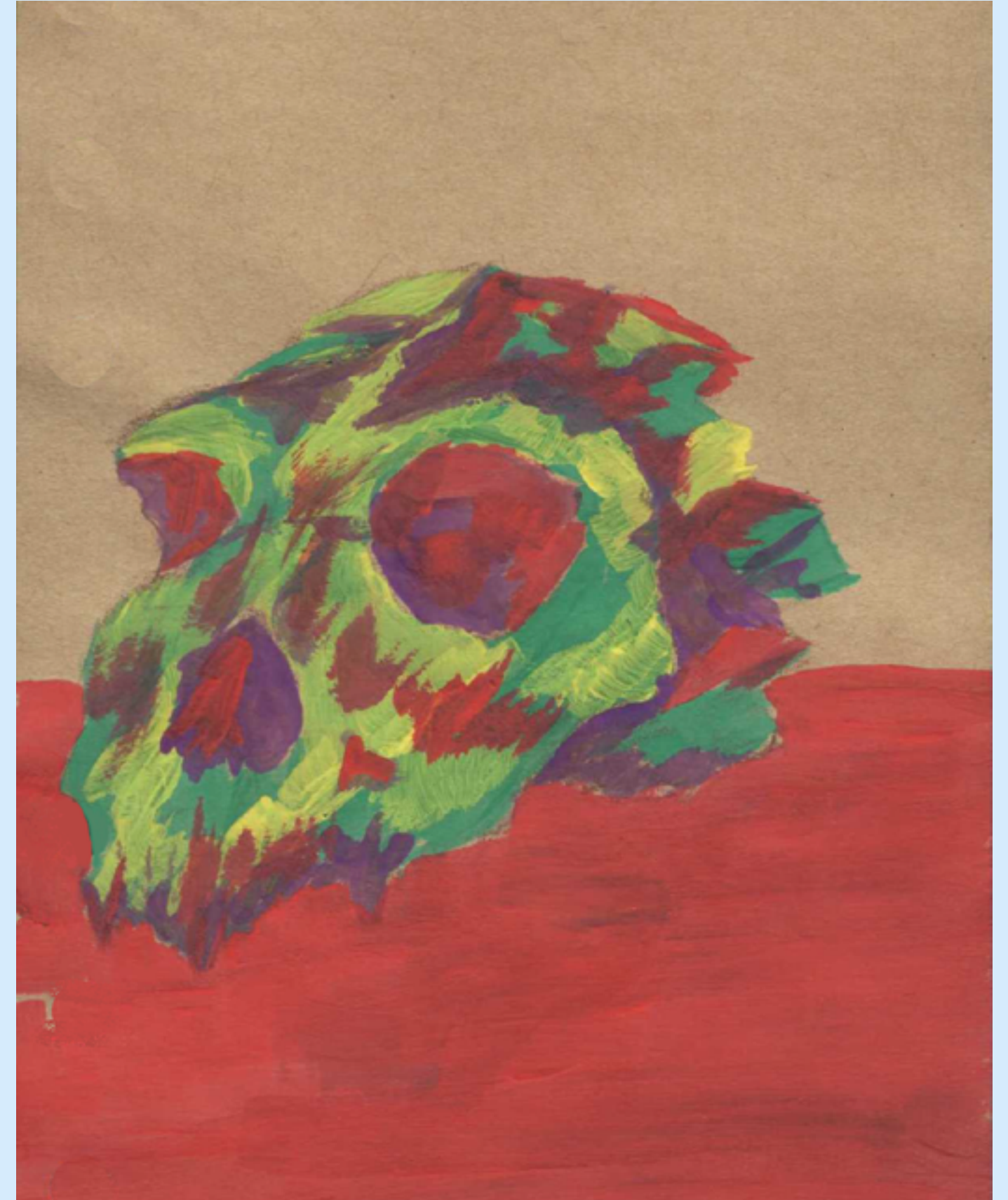
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Artwork by Liliette Quintana





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Artwork by Ray Atlas

Hugo de Vries's Mutation Theory: the Biosocial Implications of Evolution

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Abstract

This work explores the impact of Hugo de Vries's Mutation Theory on evolutionary thought. De Vries, a Dutch botanist and geneticist, challenged traditional Darwinian views by proposing that evolution occurs through sudden, discontinuous mutations rather than gradual processes. Analyzing his seminal works, "Intracellular Pangenesis" (1889) and "The Mutation Theory" (1901-1903), this paper examines de Vries's shift from pre-Mendelian ideas to embracing Mendel's laws. While his mutation theory has been largely disproven, de Vries's contributions revolutionized evolutionary theory, emphasizing the role of mutations in speciation. The paper also delves into the historical context, highlighting de Vries's interactions with Darwin and his evolution from a critic to a proponent of Darwinian principles. Additionally, his social vision anticipated advancements like genetically modified organisms and eugenicism, showcasing the enduring relevance of his work in both scientific and societal contexts. Despite subsequent challenges to his theories, de Vries left a lasting impact on biological research and its sociopolitical applications by promoting

quantitative analysis and empirical evidence.

Introduction

Hugo de Vries (1848-1935) was a Dutch botanist and geneticist whose work on heredity and evolution during the late 19th and early 20th centuries was heavily influenced by both Darwin and Mendel and, in turn, influenced future scientific advances (Kottler, 1979). This paper discusses de Vries's proposed theories from two of his publications: *Intracellular Pangenesis*, published in 1889, whose critical reception inspired several years of intense genetic research, and his two-volume *The Mutation Theory*, published in 1901 and 1903, his most famous work and the culmination of his experimentations (Stamhuis, 2003). De Vries envisioned evolution not as continuous and incremental, as Darwin proposed, but as sudden, discontinuous events that arose solely due to mutations (Stamhuis, 2003). He credited the source of these mutations to be from discrete hereditary units known as 'pangenes,' which are further explained in later sections. De Vries's theory of mutation as the sole driver of evolution has been largely disproved and subsequently dismissed, and his assertion that he independently

came to the same conclusions as Gregor Mendel is steeped in controversy (Kottler, 1979). However, his contributions to evolutionary theory were revolutionary and significantly aided the advancement of evolutionary theory and the instatement of rigorous scientific practice (Allen, 1969).

Background: Contemporary Evolutionary Theory

De Vries can be understood as a product of his time: his theories, while revolutionary, were both informed and constrained by the scientific knowledge available to him. The work of Charles Darwin, and later Gregor Mendel, heavily influenced de Vries's pursuit of evolutionary theory (Theunissen, 1994b). While controversial at the time, de Vries readily accepted the basic tenets of Darwin's theory of evolution: (i) that natural selection is the driving force of evolution, and (ii) that natural selection could only work on a trait that is variable, heritable, and results in differing levels of fitness in a given environment (Boero, 2015). However, a major knowledge gap in Darwin's theory, and one which interested de Vries especially, was the lack of an explanation for genetic variation; *On the Origin of Species* "predated the discovery of genetics, the backbone of modern evolutionary theory" (Boero, 2015, Abstract). Mendel's work was unknown at the time, but de Vries had "independently rediscovered" his

theory of genetics (a claim which later was met with scrutiny) (Theunissen, 1994b, p. 291). Mendel theorized that (i) the units of heredity are either dominant or recessive, with an individual carrying both a dominant and recessive allele for a trait showing only the dominant one; (ii) alleles inherited from the mother and father are transmitted to offspring randomly, or the law of segregation; and (iii) alleles for different traits are inherited independently of each other, or the law of independent assortment (Stenseth *et al.*, 2022). Mendel's work on heredity was largely unknown in de Vries's time (Kottler, 1979). As such, de Vries's alleged independent rediscovery of the laws of heredity was heralded as a groundbreaking contribution to biology (Kottler, 1979).

Darwin and de Vries met in the 1870s, which inspired de Vries to pursue a career in the study of heredity (Theunissen, 1994b). In 1919, almost two decades after he released his groundbreaking theory, de Vries published "The Present Position of the Mutation Theory," explaining the influence his predecessors had had on him and defending the validity of his theory. In de Vries's understanding, Darwin's theory of evolution claimed that "species originate by the gradual accumulation of infinitesimal, ordinarily invisible variations on account of their utility in the struggle for life" (De Vries, 1919, p. 213). The "difficulties

inherent" in this theory, which de Vries does not explain in his paper but can be interpreted as a lack of sources for this variation, led to the development of mutation theory (De Vries, 1919, p. 213). Still, de Vries largely accepted Darwin's theory as correct and posited his theory of mutation as "a logical successor to Darwin's theory, which incorporated recently acquired insights into heredity" (Theunissen, 1994b, p. 292). De Vries immensely respected Darwin's contributions to evolutionary theory and believed that his work supported, rather than refuted, the theory of natural selection.

At the time of *Intracellular Pangenesis*'s publication, de Vries's claims were met with intense criticism for neglecting the role of natural selection and ignoring continuous variation. This inspired him to devote years of his life to rigorous experimentation on the evening primrose, *Oenothera lamarckiana*, which led to a refinement of his theories in *The Mutation Theory*, about a decade later (Stamhuis, 2003). De Vries wrote in this second publication:

"In my *Intracellular Pangenesis* I expressed for the first time the idea that these factors are not the organs and the cells, which until then was the common opinion, but the properties active in the protoplasm. Since that time (1889) the hope of an experimental foundation of this propo-

sition has been the guideline of the investigations which form the subject of this work" (Stamhuis, 2003, p. 120).

Similarly to Mendel's pea plant experiments, de Vries observed phenotypic changes in successive generations of flowers (Stamhuis, 2003). He attributed these changes to sudden mutations, which eventually resulted in an entirely new species (Stamhuis, 2003).

The rediscovery of Mendel's work after de Vries publicized *Intracellular Pangenesis* in 1889 caused de Vries to change his conception of heredity (Theunissen, 1994a). The exact changes in de Vries's theory will be discussed in the next section. Still, it is important to note Mendel's influence here because of the controversy surrounding de Vries's assertion that he had already discovered Mendel's laws before reading his work (Theunissen, 1994a.). The genetic experimentation that de Vries conducted over almost thirteen years resulted in patterns of inheritance that did not exactly match Mendel's ratios. Like Mendel's pea plants, de Vries isolated and selected plants of the evening primrose, *O. lamarckiana*, with distinct phenotypic variations, and then established stable, self-pollinating lines to determine the laws of mutation. In the second-generation hybrids, de Vries discovered a 2:1 and 4:1 ratio, but not Mendel's own 3:1 (Theunissen, 1994a.). These ratios,

published in 1897, defy de Vries's claim that he had independently discovered Mendel's laws in 1896 (Theunissen, 1994a). In fact, he only discovered a 3:1 ratio in his experiments in 1900, after he claimed to have discovered Mendel's work (Theunissen, 1994a). His previous experiments had resulted in different ratios of heredity; the fact that his results so quickly changed to align with Mendel's work immediately upon his discovery of Mendelian genetics weakens de Vries's credibility. This evidence may call into question the validity of de Vries's results and his credibility, but it cannot be denied that he did bring the laws of heredity to the forefront of evolutionary theory.

Furthermore, it is helpful to consider that de Vries's interpretation of his experimental evidence was limited by his time's scientific knowledge and terminology. De Vries "had good reasons for his views" and his assumptions "offered a way of handling the available evidence that was plausible, logical, and fully acceptable according to contemporary standards" (Theunissen, 1994b). In de Vries's time, there was no explanation for the genetic cause of mutations (Nei and Nozawa, 2011). As such, he had no choice but to identify any heritable changes in phenotype as resulting from mutation (Nei and Nozawa, 2011).

De Vries's Theories

In *Intracellular Pangenesis*, de Vries outlines his pre-Mendelian views on heredity. The basic tenets of his claim have withstood the test of time: each cell in the body contains the same hereditary material, and characters are inherited independently (Stamhuis, 2003). These units of heredity, which de Vries coined as 'pangenes,' are permanently ingrained in all cells, and identical in both somatic and germline cells (Theunissen, 1994a). Any change in a pangene is automatically correlated with a change in the character that it directly represents. Pangenes can be further categorized as either active or latent, wherein only active pangenes can affect the character that they are associated with. De Vries viewed the active and latent states as temporary, rather than fixed. This idea is in stark contrast to Mendel's conception of genes as stable. In Mendelian genetics, phenotypic variation is attributed to the concept of dominant and recessive genes, meaning that it is their interactions, rather than a change in their state, that causes a change in characters. New characters could only arise from sudden pangenic mutations. De Vries theorized that pangenes can only become active if they are present in large quantities within the cell, so a new character can only be expressed as a result

of successive generations all carrying the mutated pangene in increasingly large amounts. Therefore, he viewed evolution not as continuous and incremental, but as discontinuous, abrupt, and occurring over a much shorter period (Theunissen, 1994a).

De Vries's conception of pangenes in *The Mutation Theory* is significantly different from his explanations prior, attributed both to his years of independent experimentation and his discovery of Mendel's laws (Theunissen, 1994a). In his research, de Vries observed mutant species of *O. lamarckiana*, identified by variations in petal color, leaf shape, and height, which appeared abruptly and did not display atavism as "degressive" species did (Theunissen, 1994a, pg. 239). He interpreted these sudden new species as being due to progressive mutations. Firstly, de Vries redefined active and latent pangenes from temporary to permanent states, postulating that new phenotypic variation could only result from mutation. In subsequent publications, de Vries replaced the terms active and latent with dominant and recessive, borrowing Mendel's terminology. His theory of mutation became much more complicated. Pangenes are described as discrete, hereditary particles residing in a cell's nucleus, which are transmitted from generation to generation through germ cells. Variation can be

continuous (as a result of different concentrations of pangenes through the generations) or discontinuous (as a result of mutations causing new pangenes to arise in subsequent generations) (Theunissen, 1994a). Only discontinuous variations result in new species – in short, only mutations can cause the formation of new species. He goes on to describe three different kinds of mutations: regressive, degressive, and progressive. Regressive mutations result in a loss of function, wherein active pangenes regress to a latent state, causing the absence of a previously present character (Theunissen, 1994a). Degressive mutations are "partial variations," an intermediate, partially functional state that explains the atavism he encountered in his experiments (Theunissen, 1994a, pg. 239). Finally, progressive mutations are the cause of speciation and evolution.

In short, mutation theory supposes that evolution, and specifically speciation, "proceeds by small but distinct steps," cementing de Vries's conception of evolution as discontinuous, abrupt, and short (De Vries, 1919, p. 213). Furthermore, de Vries believed that mutation events could affect entire species, leading to "a new variety or sub-species" being "formed all at once, [continuing] to breed true according to the laws of Mendelian heredity" (Bowler, 1978, p. 59). This differs from Darwin's theory of evolution as a long,

continuous process that acts on individuals rather than populations, but it does not negate the importance of natural selection for, as de Vries writes, “it is only after [a trait’s] appearance that the environment can decide about their utility” (De Vries, 1919 p. 213). Mutations “served simply as a new starting point for selection”; mutation can explain the process as being much faster than Darwin, and it can even explain the presence of “useless qualities,” or vestigial traits, as mutations that are insufficiently disadvantageous to be acted upon by natural selection (De Vries, 1919, p. 213).

Contributions to Evolutionary Theory

The Mutations Theory was widely accepted at the time of its publication, catapulting de Vries to international fame. His work “was the first experimental study of evolution in a large scale,” and a lack of similar research being conducted meant that there was no independent evidence that could meaningfully disprove his conclusions (Nei and Nozawa, 2011, p. 812). Although it was not his intention, anti-Darwinists quickly latched onto de Vries’s mutation theory as a way to deny Darwinian theory despite de Vries viewing his work as an extension, and not a refutation, of natural selection (Kottler, 1979). As new genetic evidence accumulated, however, de Vries’s mu-

tation theory was called into question. In later years, it was discovered that *O. lamarckiana* was “a permanent heterozygote,” meaning that each individual carries two different alleles for a single gene and that this state is maintained with successive generations (Nei and Nozawa, 2011, p. 812). Crossing over and genetic recombination do not occur during reproduction, so the offspring of a permanent heterozygote are therefore always heterozygous (Nei and Nozawa, 2011). Permanent heterozygotes maintain genetic diversity within a population (Nei and Nozawa, 2011). Since *O. lamarckiana* is a permanent heterozygote, any conclusions that de Vries drew from studying their reproductive habits cannot be generalized. Therefore, de Vries’s mutants, which he believed to be new species, were simply “chromosomal rearrangements derived from” this genetic state (Nei and Nozawa, 2011, p. 812). This discovery, among other advances in genetics, led to de Vries’s work being regarded “as a failed attempt to modify” evolutionary theory (Nei and Nozawa, 2011, p. 812).

Still, at least parts of de Vries’s theories are correct. His theory of discrete, heritable units that are identical in each cell but whose differential expression causes phenotypic variation, was scoffed at in his time but holds up with the modern concept of alleles as different variations of a

single gene. Later studies that sought to replicate de Vries’s experiments found that “at least one of his elementary species was a tetraploid,” which “established itself as a new species in self-fertilizing evening primrose” (Nei and Nozawa, 2011, p.813). Thus, while he was inaccurate about its evolutionary significance and much of his terminology is outdated, his basic proposal of mutation theory was correct.

Beyond ascertaining the validity of pan-genic and mutation theory, de Vries’s work had an immense impact on evolutionary theory and scientific research. His work led to significant changes in biological scientific practice in three major ways: (i) the development of testable, rather than non-testable, research questions, (ii) the collection of quantitative, rather than qualitative, data, and (iii) analytical and reductive, rather than synthetic and speculative, theories to break down complex phenomena into smaller, more easily digestible parts (Kottler, 1979). And, as Darwinism became more universally accepted, de Vries’s contribution of mathematical and genetic theory added validity to Darwin’s theory of evolution, which was mostly speculative (Kottler, 1979). De Vries offered empirical data, using quantitative analyses of the phenotypic variations he observed in his experimental populations. By quantifying these traits and analyzing the fre-

quency and types of variation within the population of *O. lamarckiana*, he provided evidence of the existence of genetic variability and proposed theories of its origin and inheritance. This supported Darwin’s notion of evolution via natural selection, adding much-needed explanations for how the genetic variation that natural selection acted on came to be. This “change in explanatory criteria” is de Vries’s most important contribution to evolutionary theory and marks a change in how biological research was conducted in the early twentieth century (Allen, 1969, p. 87). Thus, the major scientific advancements of the twentieth century to today were, in part, set in motion by de Vries’s rigorous analytical approach to research.

Finally, de Vries’s work has had lasting social impacts. Unique for his time, de Vries envisioned science as crucial for social progress, arguing that “knowledge of the laws of mutation will at some future time enable us to produce mutations artificially and at will and, in this way, to develop plants and animals with completely new characteristics,” with the goal of “[intervening] in the mechanism of species formation” (Theunissen, 1994b, p. 303). He correctly predicted and advocated for genetically modified organisms (GMOs), which today have ameliorated food insecurity. He embraced the social implications and practical implementations of his genetic research as both

“a defense of science as one of the motors of social progress and a theory about the nature of the social order” (Theunissen, 1994b, p. 310).

Biology could provide an understanding not only of the natural world but of the way that human society should be ordered (Theunissen, 1994b). A staunch eugenicist, de Vries recognized the role that social organization played in human evolution and believed that his research could help solve “the problem of human intellectual (in) equality and its consequences for education and the social order” (Theunissen, 1994b, p. 300).

He proposed to achieve this through selectively breeding humans with desirable traits to enhance the frequency of those qualities within the population, as opposed to the forceful sterilization of undesirable populations (Theunissen, 1994b). In the context of modern society, his political and ideological inclinations are a mixed bag. On the one hand, his research decreased food insecurity; on the other, he was a eugenicist who advocated for the elimination of certain traits from the population. Whether or not it is appreciated by modern society, de Vries’s legacy lives on today.

Conclusion

De Vries was constrained by the knowledge of his time and overshadowed by his successors, but he provided a stepping-off point to

major scientific advancement. De Vries did not ever claim to refute Darwin’s theory of natural selection, nor did he deny Mendel’s influence on his work. He did, however, present his theories as an enhancement to evolutionary theory. The introduction of genetic mutations explained variation in a way that Darwin could not, thereby providing further understanding of evolutionary processes. His theory of mutation contributed not only to the theory of evolution but to the field of biology and the rigor of scientific experimentation. His addition of quantitative data to support Darwin’s theory helped cement Darwinism as the backbone of modern biological science, leading to today’s advancements. Additionally, his rediscovery of Mendel’s laws helped solidify the central importance of evolution in biology. Furthermore, the application of science to social good, such as genetically modifying crops, is thanks to the contributions of de Vries. De Vries’s work, while controversial and largely disproven, was an important step in the right direction for evolutionary theory and scientific progress.

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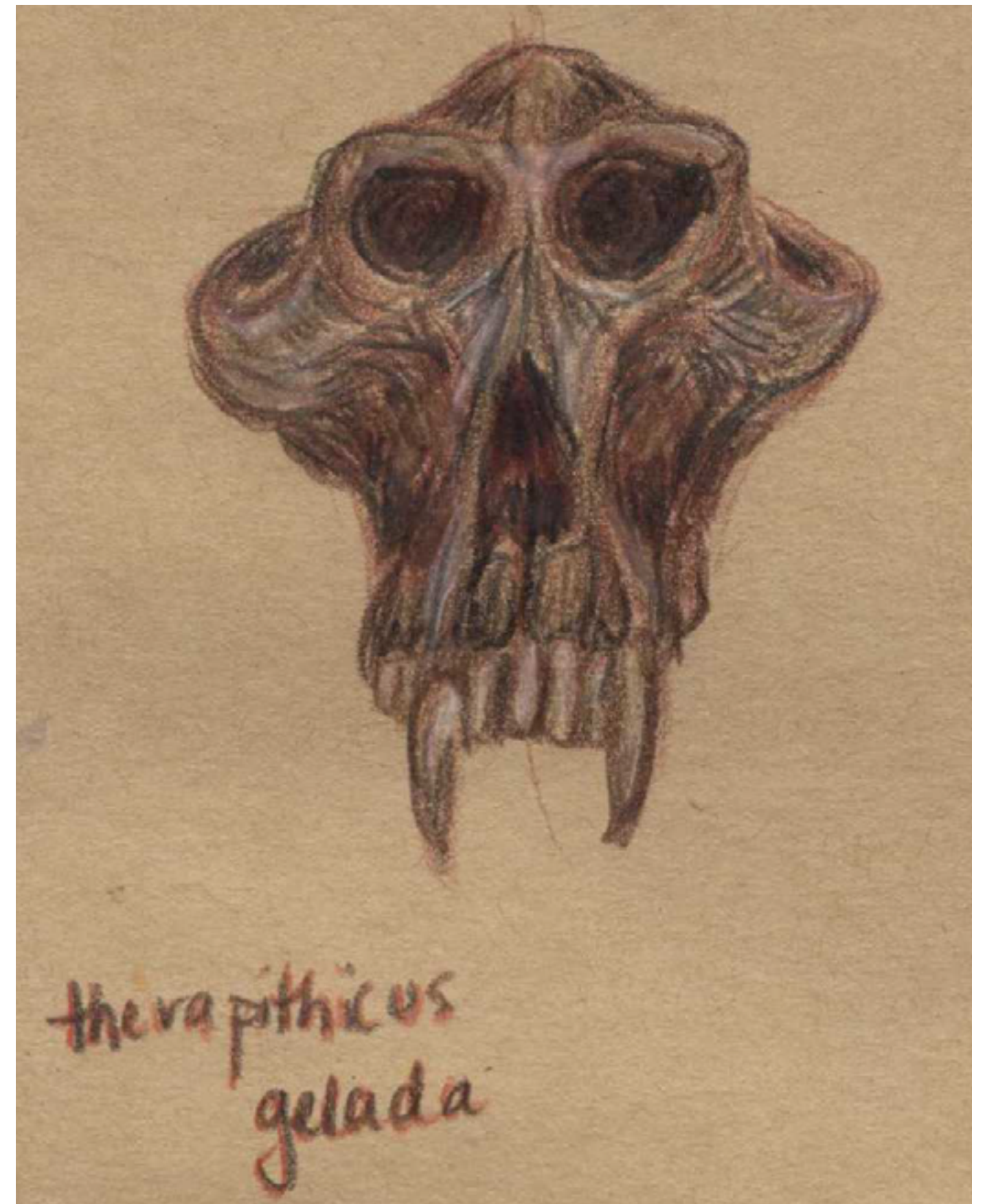
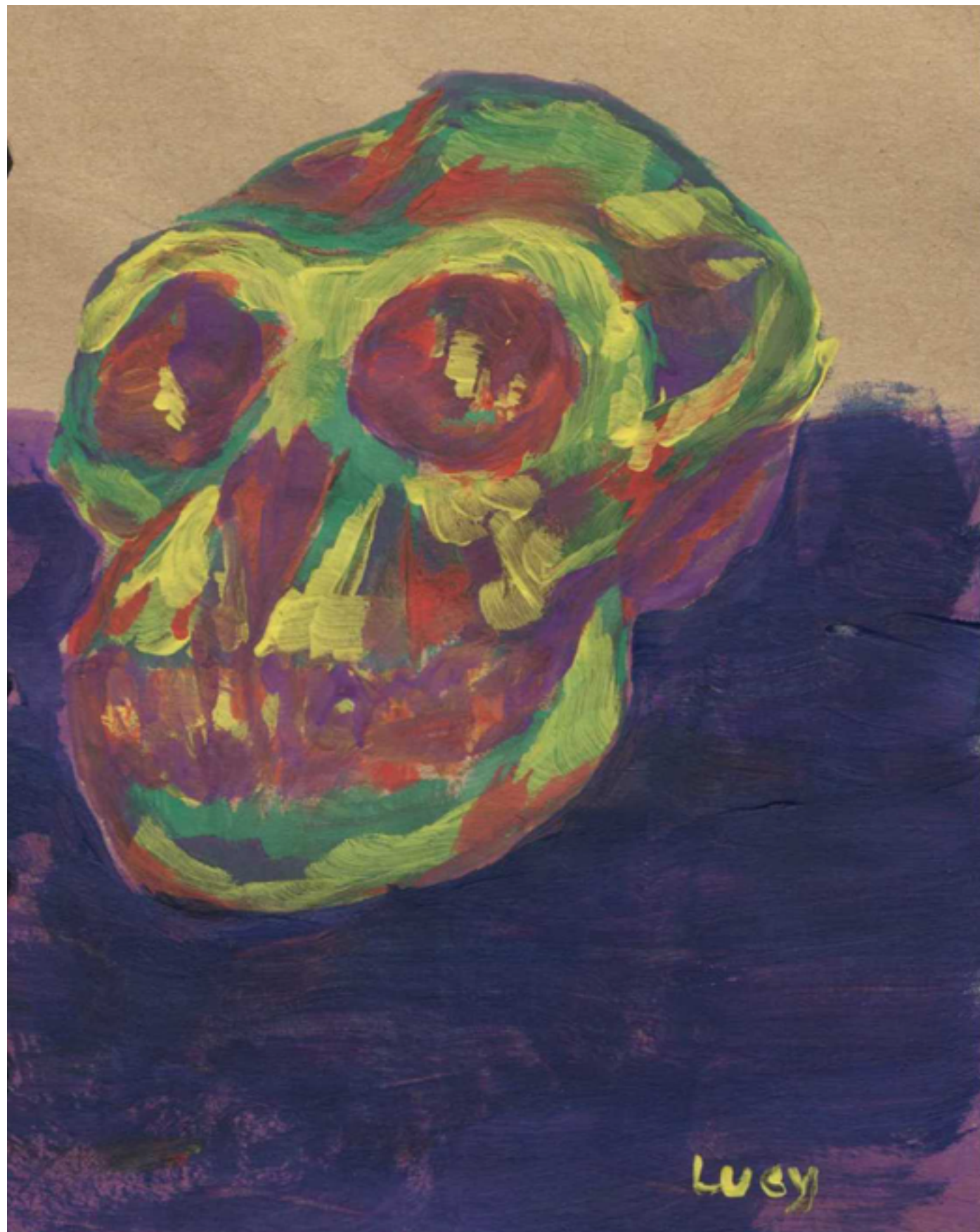
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Artwork by Liliette Quintana



Artwork by Ray Atlas

Yo Ho Ho, But No Bottle Of Lime
By Tatem Kegley

In flooded tombs of knowledge kept,
A tale of skulls that once were wept.
Among the stories, old and worn,
Lie the secrets of scurvy's mourn.

So ahoy me hearties, gather 'round,
To hear a tale on scurvy's sound.
Of bones that echo up on nigh,
Of pirates' belch and sailors' sigh.

They ate but only salted meat,
And drank on ale and rum too sweet.
No fruit nor veg did wet their taste,
Till scurvy's curse had they embraced.

Their teeth did rot, their gums did bleed,
As scurvy's taint took root with speed.
And now upon their bones of old,
The telltale signs of scurvy's told.

No lemons for these salty souls,
Their teeth fell out like scattered coals.
Their bones did map a suffering plot,
Of diets lacking, and agony bought.

Thus, raise a glass to those of yore,
Whose bones may teach forever more.
For in the depths of pirate lore,
There lies a veiled disease it bore.



Neurodivergence in the Paleolithic: Adeptness, Altruism, Art, and Autism

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Introduction

Brain disorders are a distinct phenomenon of the human condition. One such neurological disorder is Autism Spectrum Disorder (ASD), a highly heterogeneous condition characterized by differences in social-emotional reciprocity, nonverbal behaviors, difficulty in developing and maintaining relationships, and that is underscored by specialized, restricted and repetitive behavior (RRB) (American Psychiatric Association, DSM-5 Task Force, 2013). ASD is also specified as a marker for intellectual distinction, as autism has a comorbid relationship with savant syndrome, in which individuals demonstrate exceptional abilities, inconsistent with their general intellectual functioning (Daniel and Menashe, 2020). More commonly, autistic individuals are referred to as “twice exceptional,” to recognize their embodiment of both ASD and differential cognitive abilities in particular domains, such as displaying cognitive divergence i.e., exceptional cognitive advantages in one domain with profound deficits in another (Uddin, 2022, p. 510).

Regardless of where a person is on the spectrum, autistic individuals are subjected to a stigma of inferiority, low intelligence and are considered socially unfit. Their existence is often viewed as a consequence of the magnanimity of modern society, causing the stigma to be internalized among autistic individuals as norms discourage them from seeking help or accommodation (Duong and Vogel, 2022). The neurodiversity movement, a civil rights and social justice activist movement which posits that neurological disability is constructed from inequitable social, physical, and cognitive infrastructures, has made progress in destigmatizing neurodiverse conditions (Axelrod and Hooker, 2022). Still, negative perceptions of autism continue to be promoted on a global scale through media representation that emphasizes social-deficit models of ASD that reinforce perceptions of autistic people as unable to live independent and successful lives (Karaminis *et al.*, 2022). By placing ASD in an evolutionary behavioral framework, this paper aims to illuminate the legacy and significance autistic traits have had on the social and cultural

landscape of humanity.

Genes Associated with Autism

Recent studies indicate that ASD is not a contemporary phenomenon, as the genes associated with autism and related traits have a long ancestry predating the emergence of the hominin line (Castelijns *et al.*, 2020). Throughout the evolution of the human lineage, cranial capacity tripled, creating changes in brain organization, function, and intelligence (Crespi, 2016). This elevated the risk of developing more intelligent levels of functioning that operate on an imbalance of interdependent cognitive skills (Crespi, 2016). Supported by research that demonstrates that alleles associated with autism substantially overlap with alleles that are related to intelligence, ASD is increasingly viewed as a dysregulation of high intelligence—a genomic trade-off caused by the rapid increase in human cranial capacity (Crespi, 2016; Prakash and Banerjee, 2021). As brain size increased, so did the prevalence of ASD and neurodivergent people in prehistory who presented with differences in verbal and social skills, but such variability in behaviors is likely to have been valued by prehistoric societies in the Paleolithic age, beginning over two million years ago. This paper focuses on material culture and

communities that developed near the end of the Paleolithic age, specifically between 40,000 to 10,000 years ago. As ASD and twice-exceptional individuals inhabit a social-cognitive processing style that is contingent on high attention to detail and perception, stronger focus, and superior stimuli discrimination, they may have contributed to the specialization of society in prehistoric communities (Caldwell-Harris, 2021; Crespi, 2016). This contribution was perhaps due to cognitive divergence, manifesting in the archeological record as art, mechanical ability, and the fostering of collaborative morality as prehistoric groups became evermore reliant on their distinct abilities.

Although genetic evidence confirms that autistic people were present in the Paleolithic age (Spikins *et al.*, 2018a), ASD risk genes are an older class, evolving between single-celled eukaryotes from 2.7 billion years ago and bilaterians from 530 million years ago (Casanova *et al.*, 2019). As ancient genes, ASD alleles tend to be more conserved, i.e. mutation intolerant, variation intolerant, and functionally complex. Furthermore, they maintain a complex intragenic regulatory network, and produce longer amino acid chains, becoming easier targets for risk genes associated with syndromic autism, such as FMR1 (Casanova *et al.*, 2019). Conserved areas

include human accelerated regions (HARs), which are DNA sequences that fulfill two key criteria: (1) they are highly-conserved across a wide clade (e.g., primates, mammals, or vertebrates) suggesting they are loci of functional regions, and (2) they contain abnormal amounts of human-specific nucleotide changes, indicating these DNA sequences are cardinal in human evolution (St. George-Hyslop *et al.*, 2022).

Although well-conserved throughout vertebrate evolution, HARs are strikingly different in humans, as they are linked to the expansion of higher-order cognitive webworks, cognitive functioning, social behavior, brain development, and cognitive impairments, such as ASD (Bhattacharyya *et al.*, 2022). In humans, HARs indicate that the process underlying brain development is managed by gene expression operations that dictate gene regulatory elements (GREs) and their position in a spatiotemporal manner within the brain (Castelijns *et al.*, 2020). Evolutionary changes in gene activity probably emerged from alterations in DNA regulatory elements, suggesting that brain evolution is driven by differences in transcriptomes. As transcriptomes are the full set of ribonucleic acid (RNA) molecules that synthesize amino acids, variations in the evolution of the brain are likely to be mediated through protein production (Duński

and Pękowska, 2022).

Changes in GREs and HARs provide a neuronal basis for the development of autism and twice-exceptionality in the human population. Studies indicate that autism coded by single nucleotide polymorphisms (SNPs), which produce individuals without severe intellectual disability, are shown to be under positive selection (Spikins *et al.*, 2018a). ASD susceptibility genes include AUTS2, an activator of transcription and developmental regulators (Merdrignac *et al.*, 2022). AUTS2 codes for a full-length protein that localizes in cell nuclei to activate transcription genes involved in neurodevelopment (Hori *et al.*, 2021), causing dysregulation to correlate with ASD phenotypes such as developmental delays, speech and language disorders, and other intellectual disabilities (Merdrignac *et al.*, 2022). CNTNAP2 is also a long, highly conserved gene associated with ASD and is constitutive for neuronal development and synapse formation (St. George-Hyslop *et al.*, 2022). With eight HARs in its introns, loss-of-function mutations due to microdeletions or point mutations result in ASD phenotypes (St. George-Hyslop *et al.*, 2022). Overlap deletions in CNTNAP2 HARs tend to be heterozygous, further indicating that the loss of even a single copy in a HAR may be enough to affect brain function; insinuating the precarity

of normal cognitive functioning in humans (St. George-Hyslop *et al.*, 2022).

The NOTCH2NL gene family is implicated in autism, as its amplification is well-detected in ASD patients (Duński and Pękowska, 2022). The emergence of the NOTCH2NL gene family increased the activity of signaling pathways in neural stem cells, ascribing them an enhanced capacity to self-renew, which led to the expansion of the brain cortex in humans (Duński and Pękowska, 2022). The standard functioning of this gene family is essential in ensuring suitable brain size and neuronal activity, yet it confers a trade-off between an advantageous gain in brain size and intelligence and an increased susceptibility to ASD and cognitive divergence if dysregulation occurs (Duński and Pękowska, 2022). Collectively, ASD is characterized by a polygenic architecture that persists in populations despite a negative fitness effect (Mozzi *et al.*, 2017), with a heritability rate between 64% to 91% and a population level of 1-2% of adults living with the disorder (Duński and Pękowska, 2022). Although impaired by social deficits, communication problems, and RRBs, ASD confers advantageous phenotypes such as: increased attention to detail, superior observation skills, focused concentration, increased long-term memory and visual imaginative skills, greater

analytical abilities, and unique thought processes that result in innovative solutions (Prakash and Banerjee, 2021). Positive selection of these phenotypes may be due to the composition of Paleolithic communities, as the diversity of thought linked to high intelligence in autistic individuals would have provided talents crucial to survival in a range of harsh and unpredictable environments.

Behavior and Specialization

The utility of autistic traits and cognitive divergence in prehistoric societies can also be understood as a modality of Enhanced Perceptual Functioning (EPF) (Mottron *et al.*, 2006). Behaviors such as systemizing, superior low-level processing, hypersensitivity to stimuli and patterns, and an advantage in utilizing fluid intelligence may have also been accorded certain respect in the widely varying Paleolithic environments rather than viewed as compounding deficits (Spikins *et al.*, 2016). EPF distinguishes ASD as a condition of superior abilities in low-level, local perceptual processing – i.e., attending to the specific details of a stimulus without interference from top-down categorization. This is different from disorders characterized by severe deficits in global processing and connectivity –i.e., perceiving stimuli in a more

general, holistic approach that is shaped by top-down gestalts (Mottron *et al.*, 2006; Uddin, 2022). Namely, autistic people have a local processing bias that aids them in discrimination and sensory acuity, visuospatial operations, and pattern recognition compared to neurotypicals that usually default to global processing (Mottron *et al.*, 2009). Although autistic people display superiority in local processing, they are not confined to this mode of perception; their plasticity allows them to assume global processing with little reduction (Chamberlain *et al.*, 2013). This adaptability of thought and twice-exceptionality may have been essential when the northern latitudes of Paleolithic Europe became colder, as that environment would have required highly complex and reliable technology to survive, fostered by unique thought processes that cultivate innovative tools (Spikins, 2013).

The tendency of autistic people to process locally is correlated with cognitive divergence and that may have proliferated the increase of specialization in prehistoric societies (Hunt and Jaeggi, 2022; Uddin, 2022). The human brain is an organism built for pattern-seeking, perception, processing, maintenance, and recognition, all of which show enhancements in individuals with autism (Crespi, 2021). Autistic people, therefore, are gifted in recognizing

repeating patterns in stimuli, which prompts a desire to systemize: a drive to observe, analyze or construct rule-based systems that function in an “if-then” manner (Crespi, 2021; Uddin, 2022). Systemizing is also a model for explaining non-social features of autism, including specialized interests, RRBs, and a compulsion for homogeneity (Baron-Cohen *et al.*, 2009). As autistic individuals have greater sensory sensitivity and acuity, they often experience sensory overloads that are managed by reducing perception of sensory input, self-stimulation by simple repetitive motor behavior (RMB) to calm the autonomic system, or self-stimulation by increasing focus on particular patterned stimuli and the blocking of other sensory inputs (Crespi, 2021). While restricting stimuli may develop into a maladaptive phenotype, RMB's that focused on specific stimuli were likely significant in fostering unique inventions, as RMBs fostered motor tinkering that initiated trial-and-error discovery (Caldwell-Harris, 2021). Therefore, hyper-systemizing, sensory acuity, and RRBs during the Paleolithic age likely increased ASD individuals' aptitude, material expertise, and the proliferation of specialized skills analogous to the abilities presented in modern savant syndrome and twice-exceptional people. As the materials of interest in savant syndrome consist

of codes, such as the systemizing of collectible (e.g. distinguishing between stones or wood), mechanical (e.g. tool production and building), numerical (e.g. calendars and mathematics) or abstract (e.g. syntax, music, or art) systems. Expertise in these areas likely provided great utility to prehistoric societies (Motttron *et al.*, 2006; Spikins *et al.*, 2016). Through incorporation into a Paleolithic community, twice-exceptional individuals may have been a constitutive component in the development of specialized roles, creating positions where autistic individuals and associated neurodivergent traits would be valued.

Art and Autism

Being an artist was a specialized position in European Paleolithic communities where autistic individuals may have excelled. The phenomenon of art directly engages humanity's most distinct qualities: the need to express our inner emotions, to annex the unfamiliar with symbols of familiarity, and as an aid to supplement the long-established pastime of story-telling. Inhabiting this role would confer status upon ASD individuals, as cave paintings were multifunctional, perpetuating social, religious or teaching functions, integral to maintaining the bonds and the survival of communities.

Although current technology has been unable to genetically place autistic individuals as the artists of European Paleolithic cave paintings (e.g. the Chauvet Cave, Les Trois Frères, and Lascaux), the artists of primitive art demonstrate an ability to perceive anatomical details similar to art drawn by current autistic people (Folgerø *et al.*, 2021). Parietal art, which are depictions of art in caves, dated to Paleolithic Europe and art by modern autistic savants both exhibit a focus on components rather than holistic forms, displaying superimposition of images, preeminent details, and accurate visual depiction of animals from memory (Spikins *et al.*, 2018a). Paleolithic artists display a cognitive bias toward perceiving fine edges and distinct outlines; they demonstrate an extreme local processing bias that bestows upon them a natural talent for realistic depiction.

Although a local processing bias can be identified in individuals without autism, it is a cognitive tendency quite common in ASD and may be categorized as an “autistic trait.” (Spikins *et al.*, 2018a). The neuroaesthetics of cave art substantiate Paleolithic artists possessing autistic and neurodivergent traits, as the tendency to local processing allowed both ancient artists and modern autistics the ability to access lower-level information before categorization into holistic concepts, facilitating a more efficient

process of breaking down movements and depicting naturalistic images (Folgerø *et al.*, 2021). This enhanced perception results from hyper-connectivity, larger visual areas, and increased neuronal excitability inside the brain that enhances motion perception. These traits all underlie archaic *Homo sapiens* and modern autistic brains (Folgerø *et al.*, 2021). Despite a lack of genetic evidence that definitively locates archaic *Homo sapiens* with ASD as the creators of European Paleolithic parietal art, the aesthetics of European cave paintings give insight into the neuronal wiring of archaic brains that, nonetheless, materialize similarly to the cognitive divergence associated with contemporary autistic brain organization, function, and processing.

Paleolithic artists exhibit autistic traits in their ability to mentally represent three-dimensional forms and by embedding hidden figures within their art. Meta-analyses concluded that autistics show superior performance in visuospatial tests, which is present in mobiliary Paleolithic art, such as the “lion-headed man” figurine from Stadel Cave in south-west Germany, as it displays an astute understanding of three-dimensional relationships (Spikins *et al.*, 2018a). The use of pattern recognition and embedded figures is also displayed in Plaquette 662 from Montrastruc, which depicts five ibexes

in several different overlapping orientations (Spikins *et al.*, 2018a). While the illustration of the plaque clearly defines the five forms, the original is difficult to decipher, and when shown to neurotypicals, many described the art as “curving squiggles,” and were unable to view the illustrations (Spikins and Wright, 2016, A Final Twist to the Mystery, para. 3). Yet, when shown to an audience that had autism, members were able to decipher the plaquette with ease and could easily “see” the different animals arranged around it (Spikins and Wright, 2016). These studies indicate that European Upper Paleolithic art may have a relationship with traits of autism, manifesting in the methods that produced primitive art and how it can be deciphered (Spikins and Wright, 2016). While genetic studies do place ASD people present in Paleolithic societies, they have not decisively been posited as the hands behind parietal and mobiliary art; however, the archeological record reveals the neuroaesthetics behind the artwork and highlights striking similarities in brain organization and phenotypes to that of contemporary autistic people.

Addressing the Counterarguments

A principal argument against the claim that Paleolithic art displays autistic traits and

behaviors is that, by modern diagnostics, cave painters would not be autistic savants. What contemporary society has pathologized as “autism” was typical in archaic *Homo sapiens*, as they possessed enhanced visual areas of the brain, indicating an increased capacity of visual acuity and memory for visual details and motion perception (Folgerø *et al.*, 2021). Furthermore, as archaic *Homo sapiens* did not have complex categorization and higher-order cognition built into the infrastructure of their language, they had more access to raw perception. Specifically, they had a perception closer to rudimentary sensory input that is not disturbed by higher-order concepts and top-down processing (Folgerø *et al.*, 2021). This is further substantiated as feedback resonance from higher-order cognition and acquired concepts activate inhibitory systems of neurons, resulting in a compression of signals into a prototypical representation that makes it easier to incorporate it into higher cognition that fits with generalized concepts (Folgerø *et al.*, 2021). Archaic *Homo sapiens* brains were built for processing raw information, which puts higher demands on processing ability, resulting in a larger and longer endocranial shape as compared to modern humans (Folgerø *et al.*, 2021; Neubauer *et al.*, 2018). Yet, the advent of more conceptual thought structures,

externalization of knowledge, the spread of agriculture and domestication of animals, and the self-domestication of *Homo sapiens* have correlated to brain reduction in the last 20,000 years (DeSilva *et al.*, 2021; Hood, 2014). With the reduction of cranial capacity, traits of archaic *Homo sapiens* – such as enhanced visual areas, three-dimensional understanding, and memory and motion perception – would have been reduced, manifesting in a few individuals, and becoming an atypicality compared to the whole of the population.

A non-pathological cognitive style, a behavior that fell into normal variations in archaic human populations, transformed into the highly heterogeneous condition of “autism.” Although shown to have exhibited similar phenotypes to archaic humans, autism is further linked to the prehistoric genome in terms of cranial capacity, as the condition is associated with an atypically connected and overgrown brain and an abnormal excess of neurons in the prefrontal cortex (Courchesne *et al.*, 2011). Therefore, autism may be conceptualized as a primitive trait: a cognitive realm inherited by archaic *Homo sapiens*. While diagnostically not considered savant autistics by current calibers, archaic humans encompassed a vast range of autistic traits and cognitive divergence that were likely considered the

standard during the Paleolithic age.

Another conclusive case for autistic people proliferating in the Paleolithic age is that ASD is a condition of social deficits. It is underlined by dominant societal and medical trends that conceptualize autism on deficit-based models of social-cognitive and emotional processing impairments (Hanlon *et al.*, 2022). Pickard *et al.* (2011) argue that these maladaptive phenotypes would have inhibited contributions to the larger community and would not have been tolerated or incorporated into prehistoric societies. Certainly, those with autism, and even people with talent at realistic depiction, are commonly associated with three spheres of deficits: social impairment, communication impairment, and RRBs rooted in genetic influence (Robinson *et al.*, 2016). A recent study conducted by Persichetti *et al.* (2022) also indicates an underlying physiological aspect due to the decoupling of the brain that maladaptively impacts social-communication networks. Neuroimaging of ASD individuals shows the social-communication network exhibited by left laterality in autistic individuals is similar to typically developing (TD) people (Persichetti *et al.*, 2022). Yet, the same network displays a significant increase in laterality in homotopic regions in the right hemisphere in autistic patients, negatively correlating with the same

verbal measure, and indicating that “reduced” laterality of the social-communication network is due to the two hemispheres being more independent than in TD individuals (Persichetti *et al.*, 2022).

Impairment of the social-communication network may be implicated in the evolution of social cognition that has led to humans developing a “theory of mind” (ToM): the ability to consider the thoughts of another person, such as guessing what they might understand or think (Spikins *et al.*, 2016). Autistic people and others with a local processing bias are usually less adept at comprehending the emotions of others, which may have been a maladaptive trait in close-knit communities. However, autistic individuals do provide a useful and logical ToM that is resistant to spontaneous judgment and social pressure (Spikins *et al.*, 2016). A willingness to monitor transgressors and counteract hierarchical posturing is highly regarded in hunter-gatherer contexts today and would have been valued in Paleolithic communities, as clearly agreed-upon rules are essential to intergroup collaboration and survival (Spikins *et al.*, 2016). This facilitation of cooperation was recognized and rewarded, as social status was ascribed to individuals who optimized group functioning, and further contributed to the division of labor and cognitive

specialization of Paleolithic communities (Hunt and Jaeggi, 2022). As such, a logic-based ToM became a valuable perceptual strategy as it facilitated long-term collaborative planning and placed constraints on cheating, bullying, and dominance, which encouraged the strong selective pressures on pro-sociality (Spikins *et al.*, 2016).

Selection for pro-sociality is further corroborated by autistic individuals' incorporation into society, as this would have widened the human ecological niche (Hunt and Jaeggi, 2022). By increasing the environmental heterogeneity and individual specialization that creates a more cognitively diverse population, competition between community members is reduced. As a result, they developed a "pool of competence," allowing for heightened cooperation that increased the group's overall fitness and resistance to extinction (Hunt and Jaeggi, 2022).

Altruism and Adeptness

Evidence of autistic twice-exceptional individuals in Paleolithic societies is established through the archeological record that exhibits altruism and caring for people with disabilities. Rather than archaic communities lacking the moral or tolerable capacity to tend to atypical

people, selection for prosocial behaviors occurred at least 1.5 million years ago, and complex social dynamics reminiscent of modern hunter-gatherers associated with collaborative morality became identifiable by 100,000 years ago (Spikins, 2015). Collaborative morality in archaic societies promoted equitable dynamics. This formed the basis of human evolutionary success, created communities that protected those with deficits in ToM abilities from exploitation, shifted focus to the large-scale group by valuing judgments and contributions to group wellbeing, and socially buffered vulnerabilities in resources and abilities (Spikins *et al.*, 2016). Such altruism fostered respect and appreciation through the emergence of specialized abilities. This was true even when associated with deficits, as new strategies of pro-sociality led to a broader diversity of human personalities incorporated into the community (Spikins *et al.*, 2016). Therefore, collaborative morality allowed for the creation of specialized niches that were occupied by individuals with certain talents, particularly people with ASD or associated traits (Spikins *et al.*, 2016). Their enhanced perceptual abilities and superior memory provided advantages in domains such as gathering plant resources, discrimination of animal and food properties, and acute motion perception was likely significant in the success of

hunting prey (Spikins *et al.*, 2016).

ASD individuals' tendency to systemize and engage in RRBs were also advantageous in technical skills and mechanical thinking. Concentration on the utility of material culture likely contributed to complex tool technologies, a distinct, but significant evolutionary influence on human development comparable to social and informational networking (Spikins *et al.*, 2018b). Technological skills were valued differently depending on the ecological environment, as mid to high latitudes were at times associated with unpredictable weather and resource availability required complex technology that focused on a high level of standardization, maintainability, and reliability (Spikins *et al.*, 2018b). For instance, in Upper Paleolithic Europe, the archeological record uncovers the appearance of highly experienced lithic specialists, a role that was well suited to those with autistic traits (Spikins *et al.*, 2018b). Specialization may have also contributed to the positive selection of ASD genes on the expansion of cognitive diversity, effectively creating an evolved social niche for autistic twice-exceptional individuals (Spikins *et al.*, 2018b). This evidence indicates the debilitating condition of ASD in modern society is manufactured by contemporary institutions that are unaccommodating to natural variations in

behavior (Axelrod and Hooker, 2022). Instead of an inherent disability, ASD and neurodivergence are only considered disorders because of the current socio-cultural context that has excluded them from the mainstream ecological niche, creating social difficulties associated with autism that are relational rather than a consequence of innate social impairments (Karaminis *et al.*, 2022).

Conclusion

ASD is a condition defined by a highly heterogeneous set of factors including behavioral phenotypes, genetic etiology, socio environmental conditions, and cognitive functioning. While autistic individuals with enhanced abilities demonstrate twice-exceptionality, they also exhibit cognitive divergence that confers profound deficits making functioning in modern society difficult (Uddin, 2022). The stigma surrounding neurodivergence is compounded by the narrow socio-cultural context that denies recognition of experiences that deviate from the niche neurotypical norm (Hanlon *et al.*, 2022; Karaminis *et al.*, 2022). Grounding these phenomena are the current medical and social spheres that portray autism and neurodivergence as inherent deficits, with the media further reinforcing stereotypical views

that inhibits successful integration (Karaminis *et al.*, 2022). However, ASD is variable and complex; its manifestation suggests that there are many pathways for cognitive and behavioral reality to express itself, but homogenization of the condition and the positioning of ASD to the neurotypical profile continues to perpetuate critical barriers to accommodations (Axelrod and Hooker, 2022; Hanlon *et al.*, 2022).

Nonetheless, due to the efforts of the neurodiversity movement, largely composed of autistic self-advocates that insist ASD be viewed as a valuable form of biodiversity, representation of ASD is slowly shifting from disability-based models towards a more accepting and holistic perspective (Karaminis *et al.*, 2022; Leadbitter *et al.*, 2021). The current stigma surrounding ASD is significant, damaging, and multifaceted, but exploration of the past presents a different perspective that locates autism and neurodivergence as intricately woven and accepted into the foundation of prehistoric communities (Thompson-Hodgetts *et al.*, 2020). The neurodiversity movement, therefore, may be further aided by looking at Paleolithic societies, as evidence indicates that neurodivergence brought positive advantages in artistic, technological, and pro-social development. By contemplating the role of neurodivergence in

Paleolithic societies, the neurodiversity movement may advance the goal of deconstructing barriers to the social integration of ASD and cognitively diverse people, serving as a basis of a new conceptualization of a society that recognizes, accepts, and celebrates diverse talent into its foundation.

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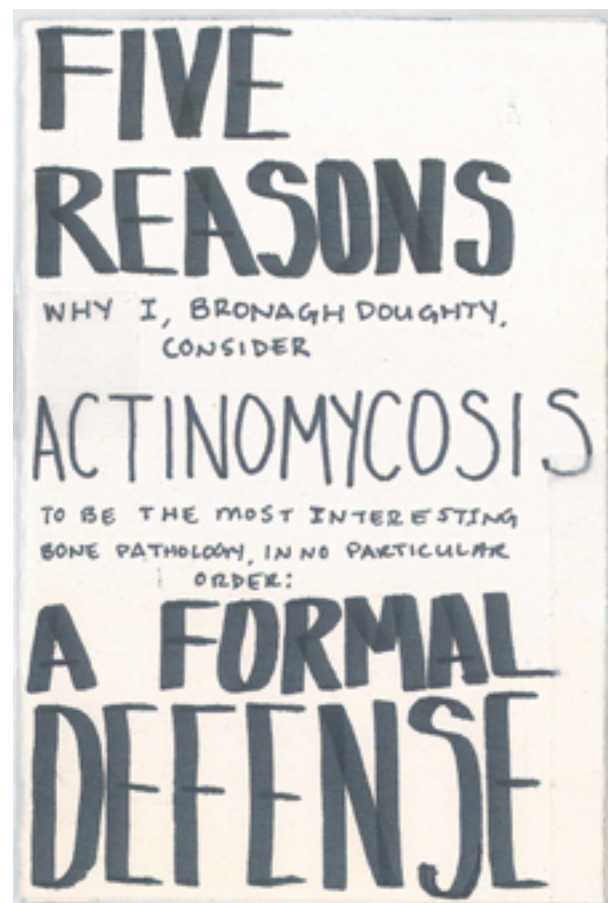
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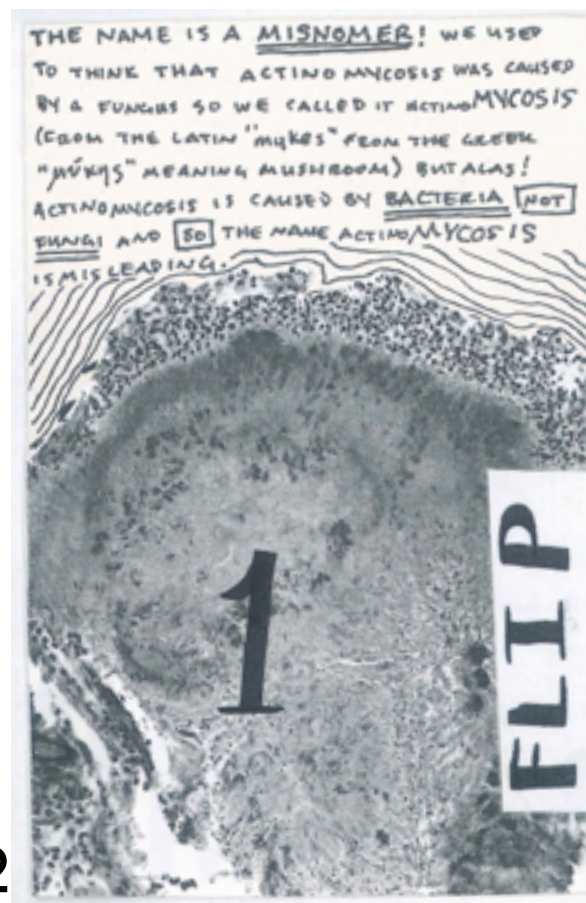
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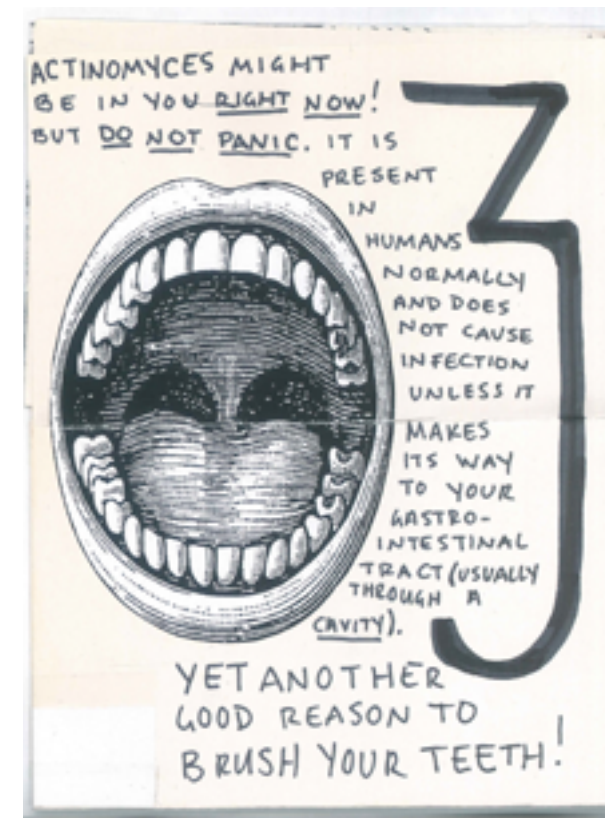


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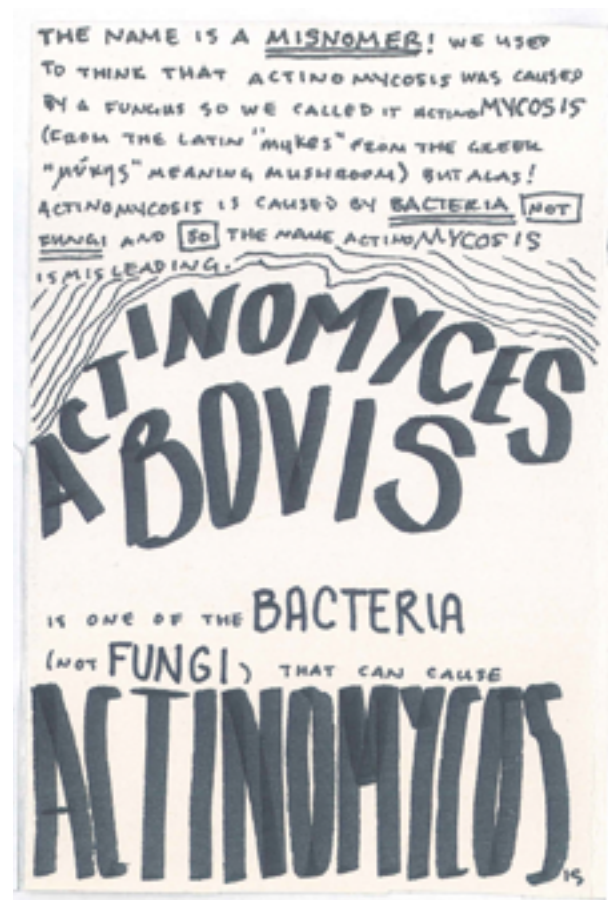
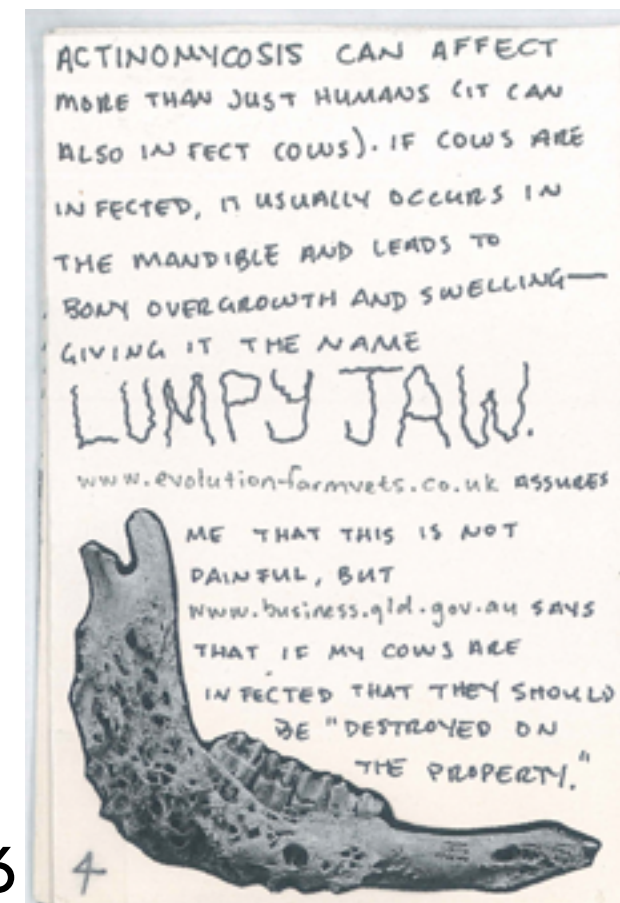
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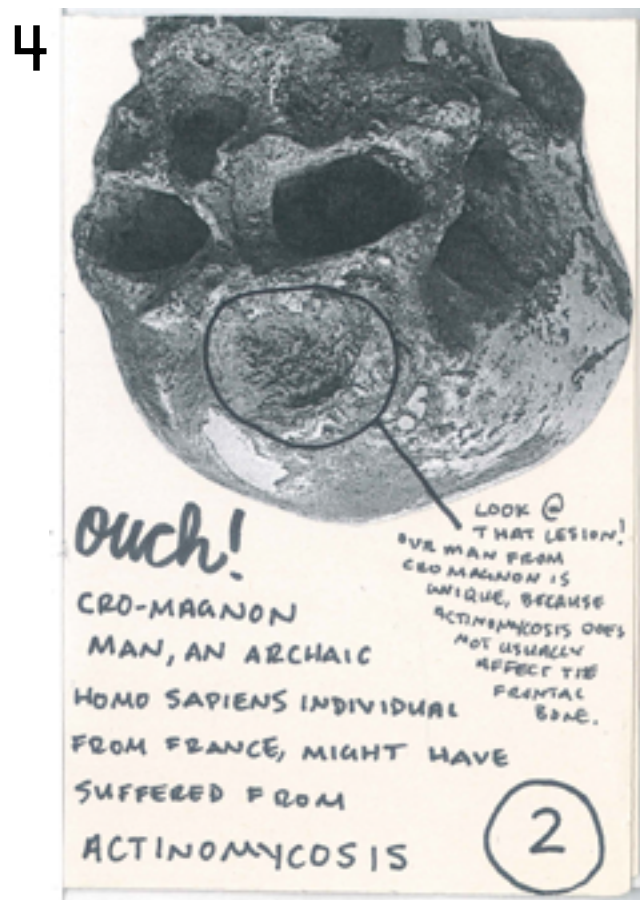


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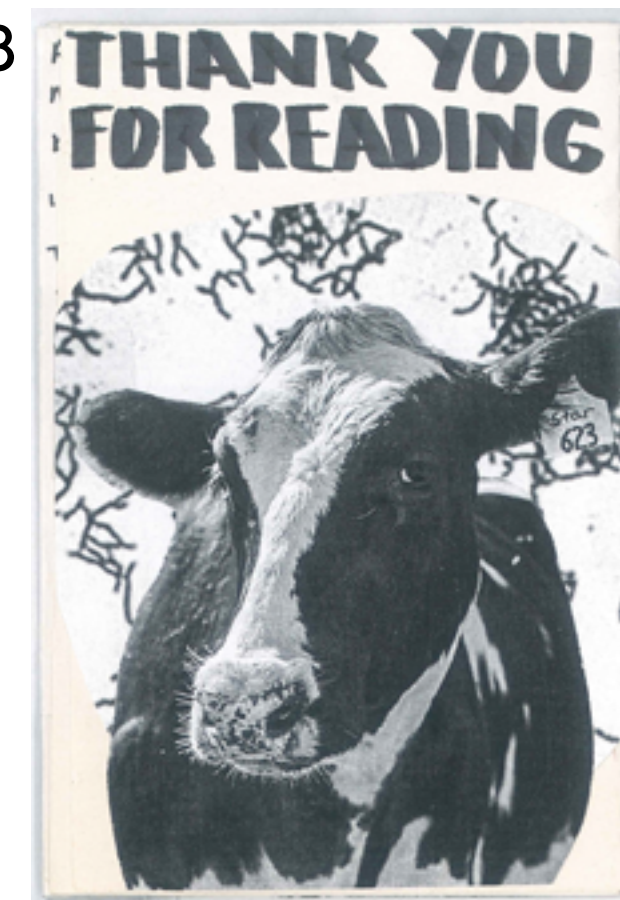


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The effect of animal protein versus plant protein on age at menopause, menarche, and childhood development

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Introduction

Albert Einstein may have been just as much an evolutionary biologist as he was a physicist. He once said that “it is my view that a vegetarian manner of living by its purely physical effect on the human temperament would most beneficially influence the lot of mankind” (Einstein, 2011, p.453).

He was absolutely right.

This paper explores the impact of protein on specific moments of life history and general life history patterns among populations – specifically, the effects of animal protein versus plant protein on age at menopause, age at menarche, and childhood development. Life history patterns are important for determining whether a population is experiencing abnormal growth or development, especially when inspecting general trends in a population undergoing rapid change.

Human diet is one such element of rapid change. Over the last few centuries, food has become increasingly processed and over-sweetened (Adams *et al.*, 2020). One change becoming ever more popular is an intentional shift toward a plant-based diet,

either vegetarian or vegan. These diets avoid animal products, including meats, poultry, and fish, often using plant-based sources of protein as replacements (Abe-Inge *et al.*, 2024).

This paper explores the potential for plant-based proteins to shift life history variables in individuals. Shifts in life history, such as earlier age at menarche or menopause, can have real implications on health and wellbeing (Boutot *et al.*, 2017; Duan *et al.*, 2020; Faubion *et al.*, 2015; Misha *et al.*, 2017). This paper’s main focus is to determine the ramifications of a shift toward plant-based proteins in diets at the individual and population level, if any exist.

This paper concludes that there are clear associations between the timing of puberty, menarche, and menopause, and the type of protein consumed during infancy and early childhood. While animal protein consumption during this period is associated with early pubertal, menarchal, and menopausal timing, plant protein consumption is associated with a delay in these three life history benchmarks. Protein consumption may simply be a correlation, however, and not a causation of the observed life history alterations. Nonetheless, there are

numerous benefits to a plant-based diet, and plant-based proteins should be a component of all diets, when possible.

Background

A recent study found that many Americans are shifting to plant-based diets due to their deemed beneficial effects on health (Storz *et al.*, 2021). The lack of animal products in a plant-based diet raises the concern that important nutrients might be inaccessible for the consumer (Mariotti and Gardner, 2019). Although plant proteins are present, they are not necessarily as effective as animal proteins (Lim *et al.*, 2021).

The quality of a protein source is determined by its amount of protein, in addition to its vitamins and minerals (Katz *et al.*, 2019). Animal proteins (beef, poultry, pork, eggs, etc.) are often referred to as complete proteins because they contain all of the essential amino acids (EAAs) (Lim *et al.*, 2021). Animal proteins also have vitamins and minerals that are otherwise difficult to obtain through diet, such as vitamin B-12 (Obeid *et al.*, 2019). Plant proteins, on the other hand, (legumes, beans, grains, soy, etc.) are considered lower quality because they only contain certain EAAs, however, they have much more fiber and antioxidants than animal proteins (Lim *et al.*, 2021).

The main function of protein in the body is the growth and repair of muscles. Lim *et al.* (2021) found that animal protein proved more advantageous for increasing lean muscle mass than plant protein, especially in young adults; they also determined that protein source has no effect on the maximum strength of muscles. Overall, the authors found no significant correlation between the physical structure of muscles and the use of plant or animal protein.

Further, Mariotti and Gardner (2019) found that vegetarian diets allow persons to obtain an excessive amount of protein and amino acids. Although some vegans may have difficulty achieving the recommended dietary quantities, the risk of lacking protein is moderate. They also argued that protein in a diet, regardless of source, is highly dependent upon plant-based protein consumption. They called for the modernization of opinions and definitions relating to protein quality because simply defining quality based on EAA content is irrelevant. If a plant-based diet is varied enough, the individual can easily ingest all EAAs necessary for high-quality health (Mariotti and Gardner, 2019).

Whether animal or plant protein is best for health continues to be a matter of debate. Comparing the effects of animal and plant protein on life history can contribute to understanding the best balance between dietary

protein and normal growth and development for both the scientific community and the general population.

Some researchers (Boutot *et al.*, 2017) propose a link between dietary intake and the onset of menopause. Boutot *et al.* (2017) conducted a study on dietary protein intake and early menopause in the Nurses' Health Study (NHS) II. Their data comprised self-reported protein intake from 1991 through 2011 using the Harvard Food Frequency Questionnaire (FFQ). In a cohort of 85,682 American women, 2,041 experienced early menopause, defined as the cessation of ovarian function before 45 years of age (Boutot *et al.*, 2017). After adjusting for all covariates, including alcohol consumption, body mass index (BMI), smoking, and months of breastfeeding, the study found that women in the highest quintile of vegetable-protein consumption (vegetable protein accounted for 6.5% of their daily caloric intake) had a 16% decreased risk for early menopause than women in the lowest quintile (vegetable protein accounted for only 4% of daily calories intake). When examining which foods had the greatest impact on early menopause, they also determined that having one serving (servings as defined by the FFQ) of red meat per day increased risk by 12%.

If age at menopause is associated with consumption of plant protein, then it is likely that

age at menarche is as well, as early menarche is a risk factor for early menopause (Mishra *et al.*, 2017). The consumption of milk, a popular form of animal protein, has been a subject of discussion for child development and early menarche. A 2011 review determined there is evidence early menarche is associated with increased milk intake among girls ages 9 to 12 years (Wiley, 2011). Milk products containing a higher percentage of dairy fat were less likely to be associated with an increased risk of early menarche.

Duan *et al.* (2020) assessed age at menarche across 1,118 6-13 year-old Chinese girls in the China Health and Nutrition Survey (CHNS). They specifically assessed age at menarche based on the consumption (or lack thereof) of the "modern dietary pattern," which they defined to be a higher consumption of fast foods, milk, fruits, eggs, and low consumption of vegetables and grains (Duan *et al.*, 2020, p.4). They did find that the highest quartile of modern diet pattern, after adjustments for confounding factors, had a 33% increased risk of early menarche. They concluded that the childhood consumption of the modern dietary pattern was associated with earlier menarche, but they found that there is no significant risk of earlier menarche with an animal-based diet.

In a study of 241 6-18 year-old Iranian girls who had not reached menarche yet, Moslehi *et al.* (2021) found a higher risk of achieving menarche with animal protein and a decreased risk of achieving menarche with plant protein. More specifically, they found a loose relationship between achieving menarche and poultry and low-fat dairy consumption, reinforcing the results from Wiley (2011) that low-fat milk is more likely to increase the risk of early menarche. Although Moslehi *et al.* (2021) did not find an explicit connection between early menarche and animal protein, corroborating the results from Duan *et al.* (2020), they found an explicit decrease in risk with plant protein consumption.

If animal protein consumption during childhood causes a change in the timing of menarche, it may also affect the timing of puberty. Günther *et al.* (2010) conducted a study based on the Dortmund Nutritional and Anthropometric Longitudinally Designed (DONALD) Study, which had a sample of 112 German children (57 girls and 55 boys). They determined that based on the timing of protein consumption, age at take-off (ATO) of the pubertal growth spurt differed by diet. Specifically, Günther *et al.* (2010, p.565) suggest that protein consumption in young children might be influenced by "critical windows" that affect pubertal timing later on. One such "critical

window" they identify is ages 3-4 years, in which they determine vegetable protein consumption affects ATO, causing it to be significantly later in groups who consume higher amounts of vegetable protein from ages 3-4 years. They also find in the critical window of 5-6 years of age that plant protein intake delays ATO.

Günther *et al.* (2010) also speculated that the effect of animal protein on ATO timing could be due primarily to dairy consumption. They found evidence of early ATO with cow milk and other dairy product consumption at ages 5-6, however, in this same age group, they did not find evidence that meat consumption caused an earlier ATO. Thus, they concluded that in the critical windows of 3-4 and 5-6 years, plant protein consumption significantly delayed ATO. In only the critical window of 5-6 years, consumption of animal protein caused earlier ATO and menarche/voice break.

Finally, the effects of protein consumption on childhood growth and development align with the associations of protein sources on timing of menopause, menarche, initiation of the pubertal growth spurt. Thorisdottir *et al.* (2014) determined that high animal protein consumption during infancy caused accelerated growth during childhood among 199 Icelandic children. These results show that animal protein intake and plant protein intake are associated with similar trends

throughout different stages of human life history.

Discussion

There are consistent assumptions and shortcomings in the preceding research. Most of the research grouped different types of animal protein together into one category, and assumed that general animal protein was a sufficiently specific factor to test for in their studies (Günther *et al.*, 2010; Lim *et al.*, 2021). Moslehi *et al.* (2021) did well at isolating for protein types, concluding that specifically poultry and low-fat dairy were particularly associated with achieving menarche.

As confirmed by this result from Moslehi *et al.* (2021), the differences between types of animal proteins are large enough to cause significant shifts in the data. In reality, there is such a diversity among types of animal proteins and types of animal-based diets that merely categorizing all animal-based proteins as one factor is limiting. Taking a closer look at the types of animal proteins, such as dairy, poultry, meat, and fish, would be helpful in future research to pinpoint the potential causes for early maturation and subsequently, possible prevention tactics. Animal proteins are distinct from plant proteins, however, and so assuming

that all types of animal proteins have similar effects on human life history is useful for finding general trends in the data.

It is important to recognize potential limitations of the data presented in this paper. Some studies have fairly small sample sizes, and most of the data was obtained from self-reports and surveys, including the Harvard FFQ, CHNS, and DONALD studies. Günther *et al.* (2010), for example, mentioned that weaknesses of their study included self-reporting, high socioeconomic status of participants, and small sample size; they also mentioned, however, one strength of their study is that in a subsample of their study population, they were able to take urinary nitrogen measurements which were used to confirm protein intake, and they also controlled for confounders. All other studies mentioned in this paper controlled for confounding variables, such as socioeconomic status, through statistical analyses. Despite these attempts, results from observational studies are almost impossible to separate from their confounding variables, and overadjustment bias can also become an issue if overcorrections are made (Villamor and Jansen, 2016).

This paper attempts to overcome these potential biases by pooling a wide-ranging set of studies. A strength of all of the data sources collected for this paper is that the

study populations are from variable geographic locations, including the United States, China, Iran, and Iceland, therefore ameliorating the problem of smaller study populations and reliance on self-reported data. For example, it makes biases such as socioeconomic status, which might be present in the study by Günther *et al.* (2010) but not Duan *et al.* (2020), less impactful. This controls for similar biases that might exist in one specific population, and it helps to prove that there is a general trend in animal proteins causing earlier life history timing across a variety of populations.

Another element of complication is the age at protein consumption. High animal protein consumption during infancy and early childhood has been proven to increase childhood body mass index (BMI)¹ (Thorisdottir *et al.*, 2014). Obesity (BMI > 30.0) is linked with early ATO (Body Mass Index... [updated n.d.]; Günther *et al.*, 2010). This suggests that protein intake at certain ages may be a confounding factor that is associated with alterations in life history timing. Thus, further research is required to establish whether age-based or source-based protein consumption is most highly associated with alterations in life history variable timing.

Another limitation of many studies, including Thorisdottir *et al.* (2014), Boutot *et al.* (2017), and Günther *et al.* (2010), is that they do not take into account the ways in which animal proteins are prepared and/or processed. The development of fast food chains and large-scale industrial meat producers has led to unnatural compounds and endocrine disruptive chemicals in certain animal protein products, which could affect ATO (Villamor and Jansen, 2016).

Duan *et al.* (2020, p.4) included fast foods in their definition of the “modern diet,” which they found to increase the risk of early menarche. Notably, they found no significant increased risk of menarche with animal-based diets independently — they only saw the increase with the modern diet. This is further evidence that early ATO may not be due to animal-based proteins alone, but is rather an issue of meat production methods and processing in fast food chains, and factory-farmed, industrially produced meat. Further research into differences in life history patterns between populations who eat primarily highly processed, industrially-produced meats and populations who eat primarily farm-raised meat would help to resolve this uncertainty.

¹ It is, in itself, a potential limitation to use BMI as a measuring factor in research. It is understood that BMI is not a reliable measurement of the body composition of an individual, and thus it is not a reliable proxy for the health of an individual. It is still widely used, however, as a measurement by which researchers discern between groups of people by body composition due to its noninvasive and inexpensive means (Body Mass Index ...[updated n.d.]).

Much of this discussion has centered on complications relating to animal proteins, but there are also complications with plant proteins. Storz *et al.* (2021) found that vegetarians had lower chances of having soft drinks, ready-to-eat foods, and salty snacks in their homes, which is one example of confounding factors present here. Very few of these studies took into account dietary differences beyond general protein consumption. Total protein consumption is only a small fraction of the dietary habits of an individual, and without correcting for the other elements of their intake, results must be taken with caution.

As such, there might be confounding factors that cause results to point toward protein as the cause of observed life history alterations, such as increased fiber intake among vegetarians than omnivores (Storz *et al.*, 2021). A study on 63 pubertal female individuals conducted by de Ridder *et al.* (1991) found that fiber is more relevant when examining estrogen levels than vegetable protein is. They found that estrogen was lower in young female individuals who ate more fibrous foods. Vegetable-based proteins contain more fiber than animal proteins (Lim *et al.*, 2021). Duan *et al.* (2020, p.4) noted that the “modern diet” is marked by decreased vegetable intake. All of the data presented here could, thus, prove

that fiber is actually the main component that helps to delay ATO and menarche.

A similar conundrum surrounds the question of whether meat or dairy products are the cause of early ATO, menarche, and menopause. Thorisdottir *et al.* (2014) found that growth in early childhood and stimulation of the IGF-1 growth hormone may be induced by dairy protein consumption during infancy. However, without isolating fiber from plant protein and dairy from animal protein, it is virtually impossible to determine which factor is responsible for changes to life history. Therefore, future research should focus on isolating specific dietary factors, obtaining data from larger and more diverse sample populations, and comparing processed and unprocessed forms of animal proteins.

Summary and Conclusions

The timing of life history matters. Early ATO has been associated with hormone-related cancers, such as breast cancer, ovarian cancer, and prostate cancer (Günther *et al.*, 2010). Early menopause is associated with a multitude of issues, including osteoporosis, heart disease, and cognitive problems like Alzheimer’s and dementia (Faubion *et al.*, 2015).

Although some uncertainty remains, individuals are suggested to add plant protein

into an animal protein-rich diet to preserve life history timing (Moslehi *et al.*, 2021). Eventually, a large-scale shift toward earlier life history timing could negatively alter human growth and development and potentially be detrimental to the human species, just as Einstein so sagely predicted.

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